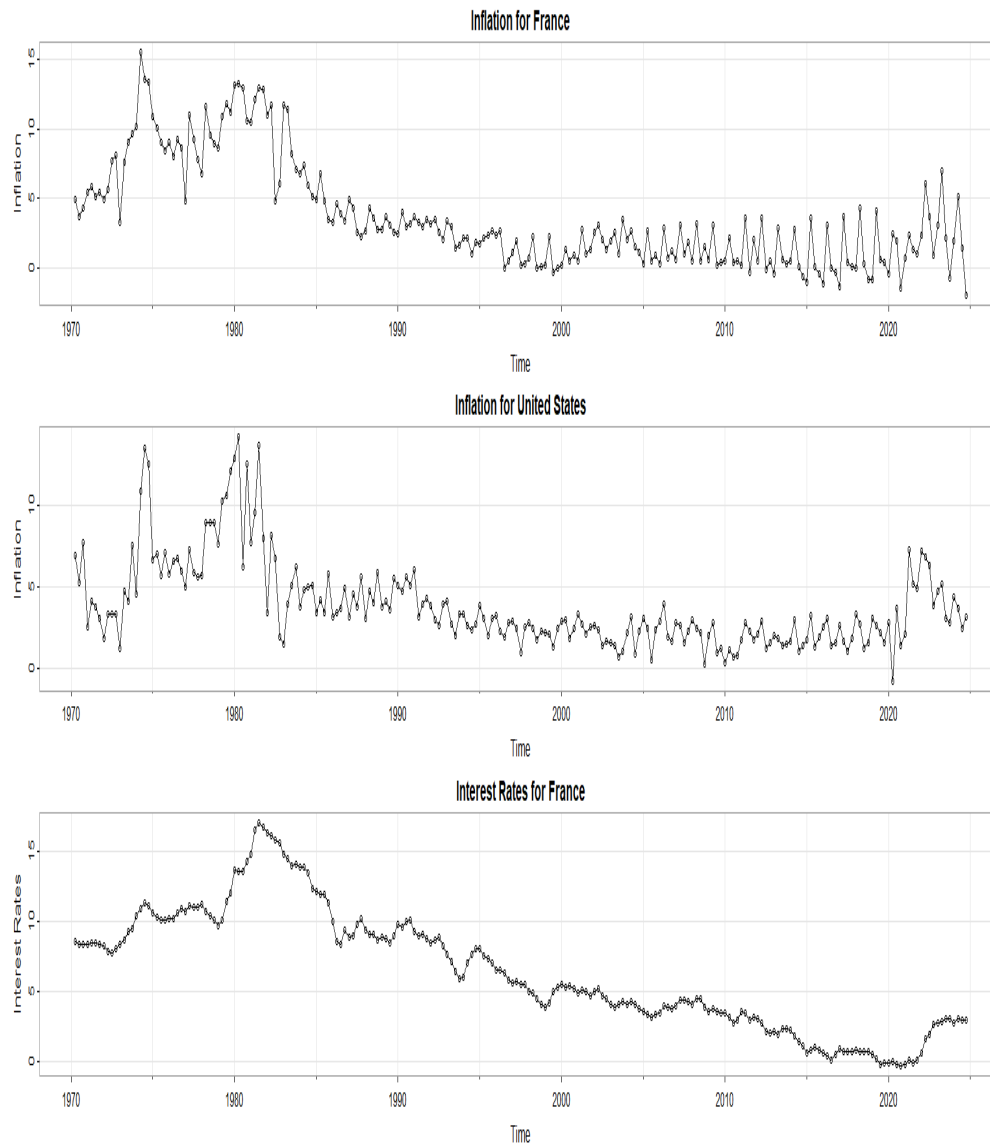


1 Choice of countries and dataset

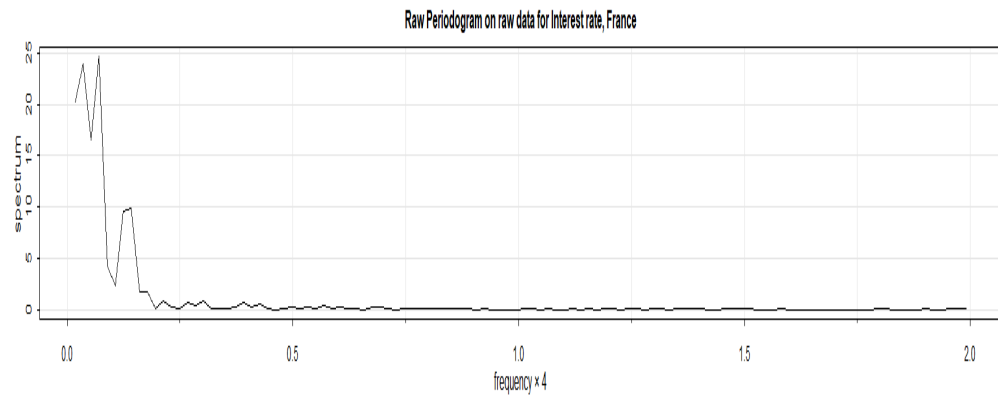
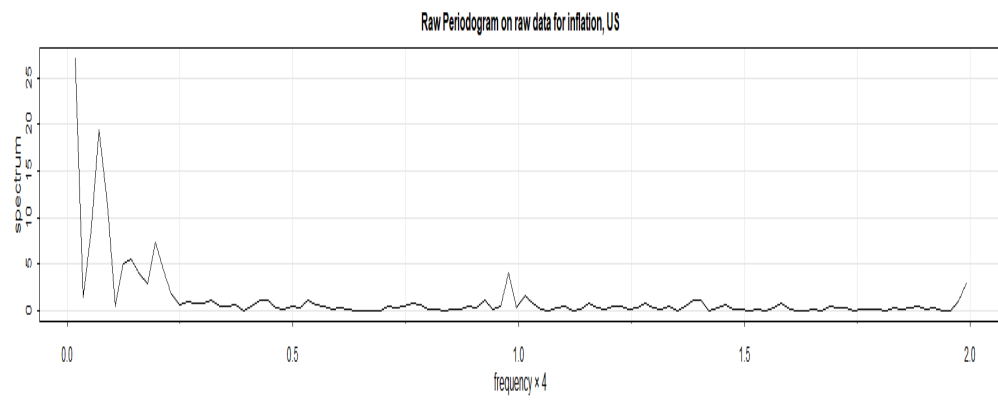
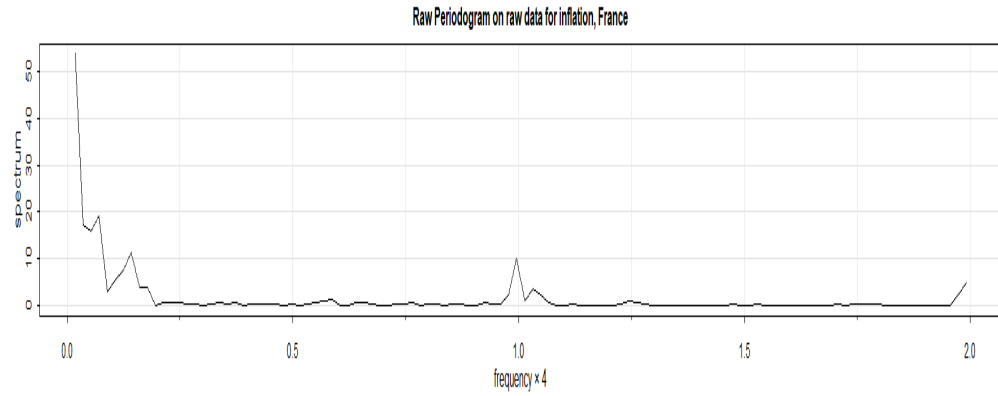
We selected France (home) and the US (foreign) because France is a large European country with relatively fewer major domestic disruptions in past decades, and France's smaller size fit models in which domestic policymakers monitor spillovers from a larger economy. We added France's 10-year government bond yield because of the well-documented link between interest rates and inflation. We use 220 quarterly available FRED observations (1970:Q1–2024:Q4) for stable results. We use Quarterly annualized inflation, computed from CPI (Index 2015=100, not Seasonally Adjusted), excluding volatile items (food and energy) for stable forecasts.

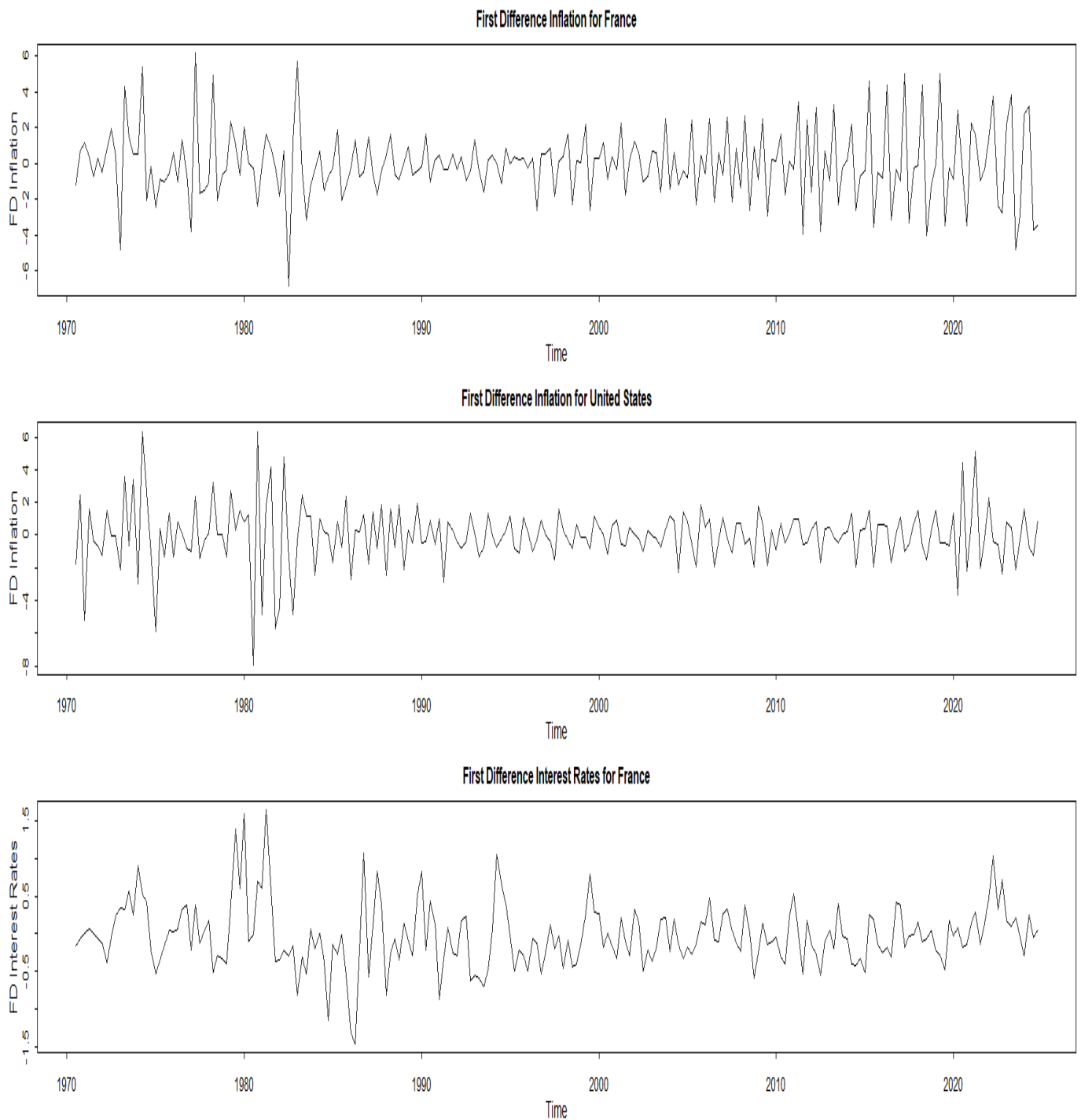
```
# Housekeeping
rm(list=ls()) # clear workspace
graphics.off() # close all graphs
library(vars) # VAR tools
library(astsa) # time series tools
setwd(dirname(rstudioapi::getActiveDocumentContext()$path)) # set wd
couleur=1 # plot color
PRICEFR=read.csv("FRACPICORQINMEI.csv",col.names=c("t","PRICEFR"))#France
CPI
PRICEUS=read.csv("USACPICORQINMEI.csv",col.names=c("t","PRICEUS"))#US CPI
IR10FR=read.csv("IRLTLT01FRQ156N.csv",col.names=c("t","IR10FR")) # France
10-year interest rate
PRICEFR=ts(PRICEFR[,2],start=c(1970,1),frequency=4)#France CPI
PRICEUS=ts(PRICEUS[,2],start=c(1957,1),frequency=4)#US CPI
IR10FR=ts(IR10FR[,2],start=c(1960,1),frequency=4)#France interest rate
INFFR=400*diff(log(PRICEFR)) # annualised quarterly inflation for France
INFUS=400*diff(log(PRICEUS)) # annualised quarterly inflation for US
Start=1970 # sample starts in 1970Q1
INFFR=window(INFFR,start=c(Start,2),end=c(2024,4)) # Start Q2 after diff
INFUS=window(INFUS,start=c(Start,2),end=c(2024,4)) # Start Q2 after diff
IR10FR=window(IR10FR,start=c(Start,2),end=c(2024,4)) # restrict sample
```

2 Graphical Analysis

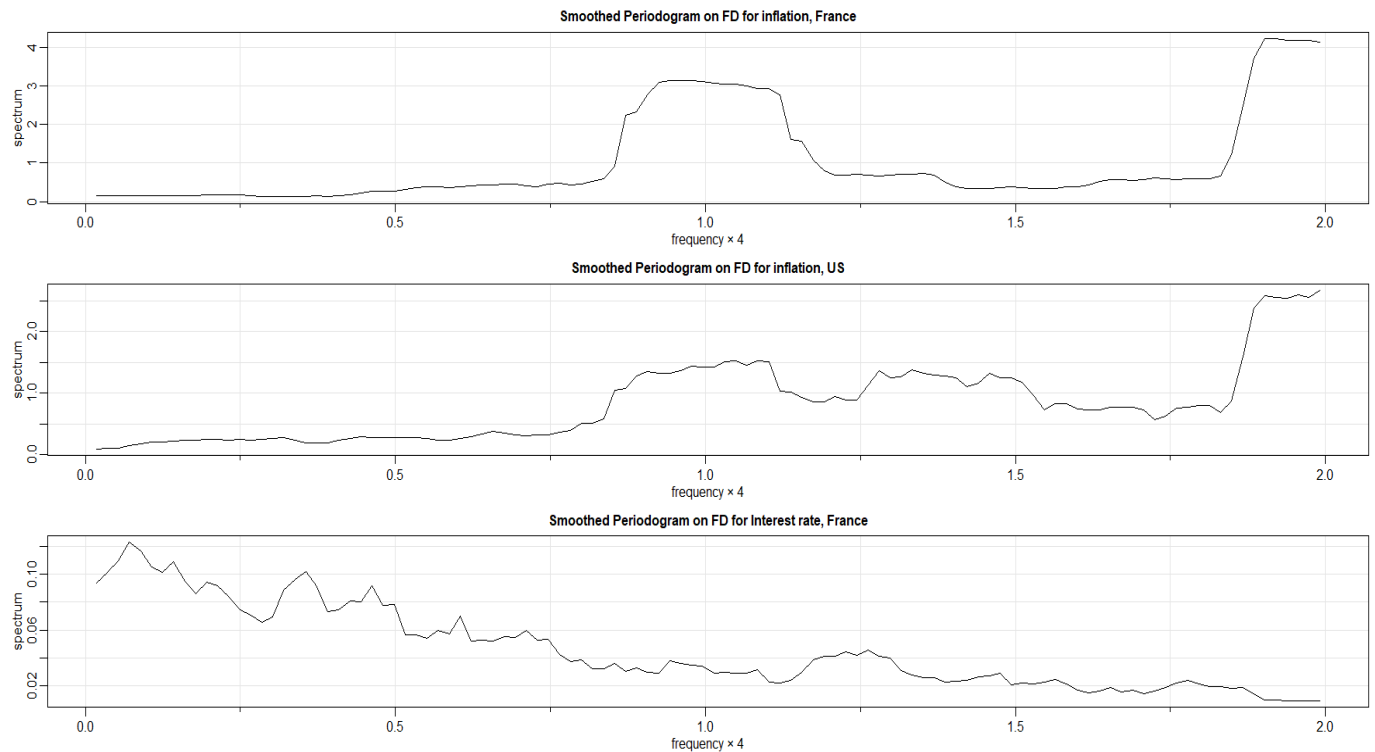


All series are persistent and do not appear stationary, which is consistent with the periodogram (next figure) showing large values near frequency zero. We suspect a structural break in inflation around 1985 in France and 1983 in the US.

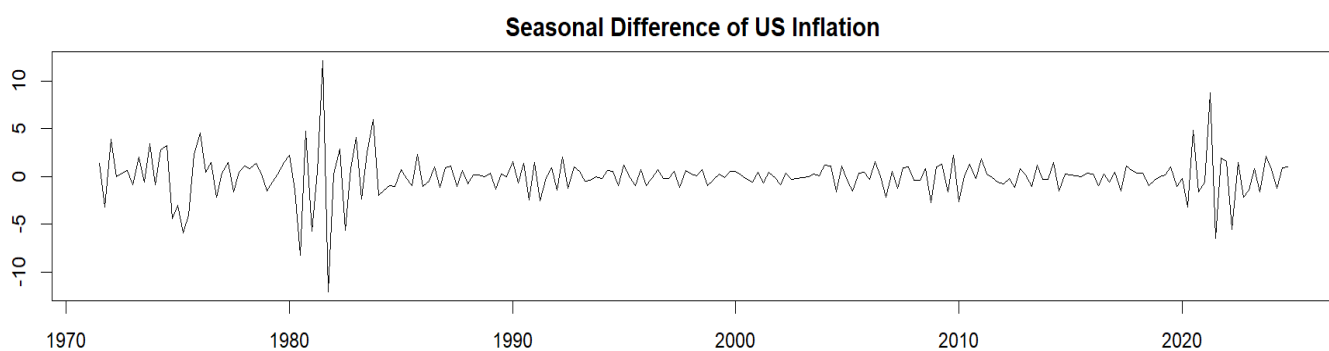
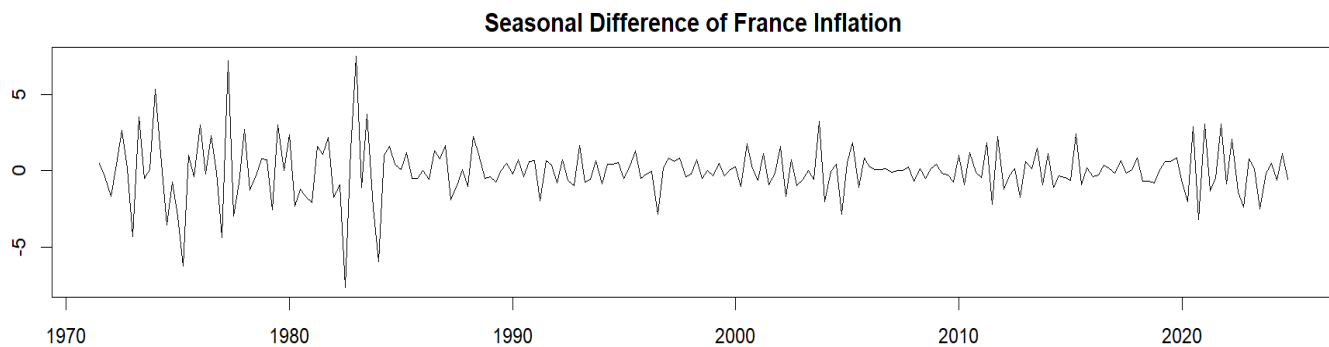




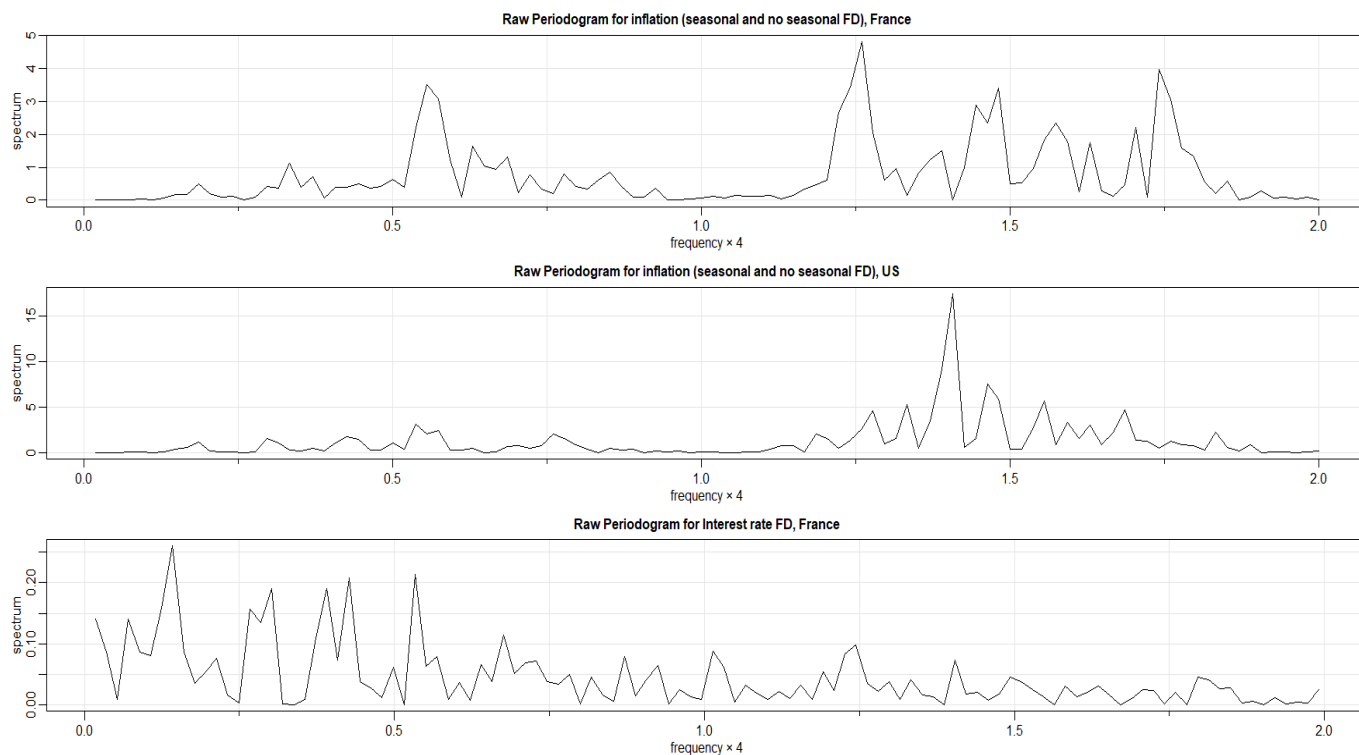
After first differencing, all series appear stationary around zero. Both inflation series show quarterly cycles.



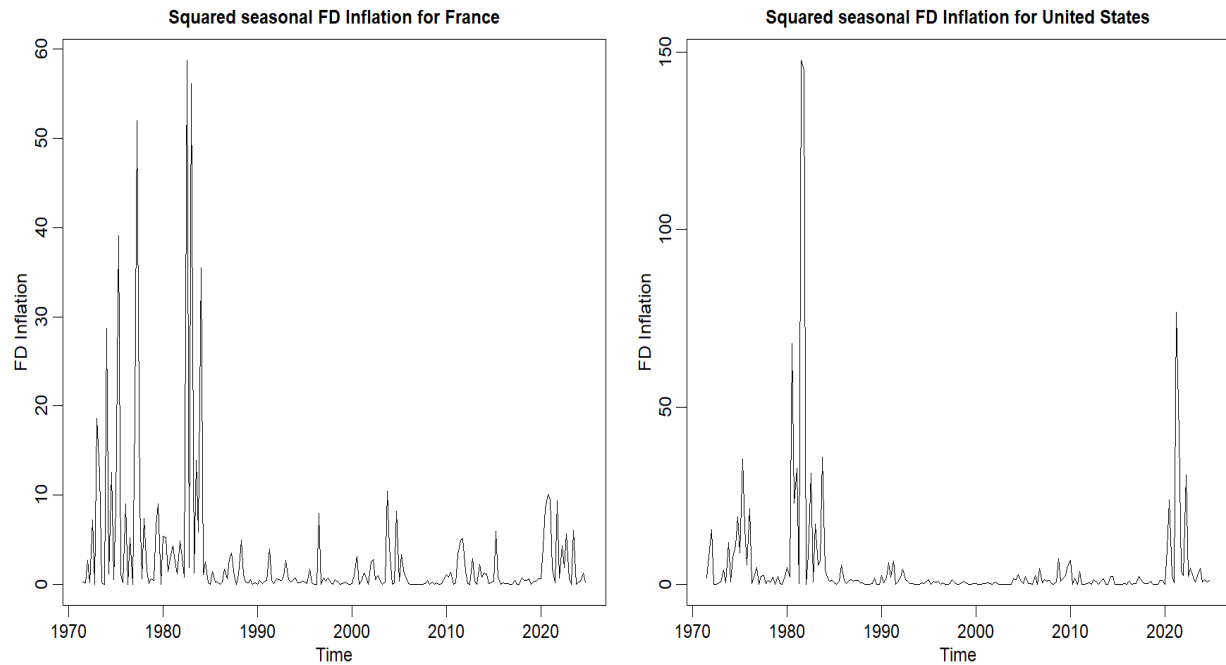
Both inflation series show annual seasonality. The interest rate shows longer cycle, with its main peak around $\text{frequency} \times 4 = 0.0625$ (64 quarters). Because such a seasonal differencing would remove substantial information, we retain the interest rate without seasonal differencing and interpret this pattern as a possible additional sign of long-run relation with the other variables.



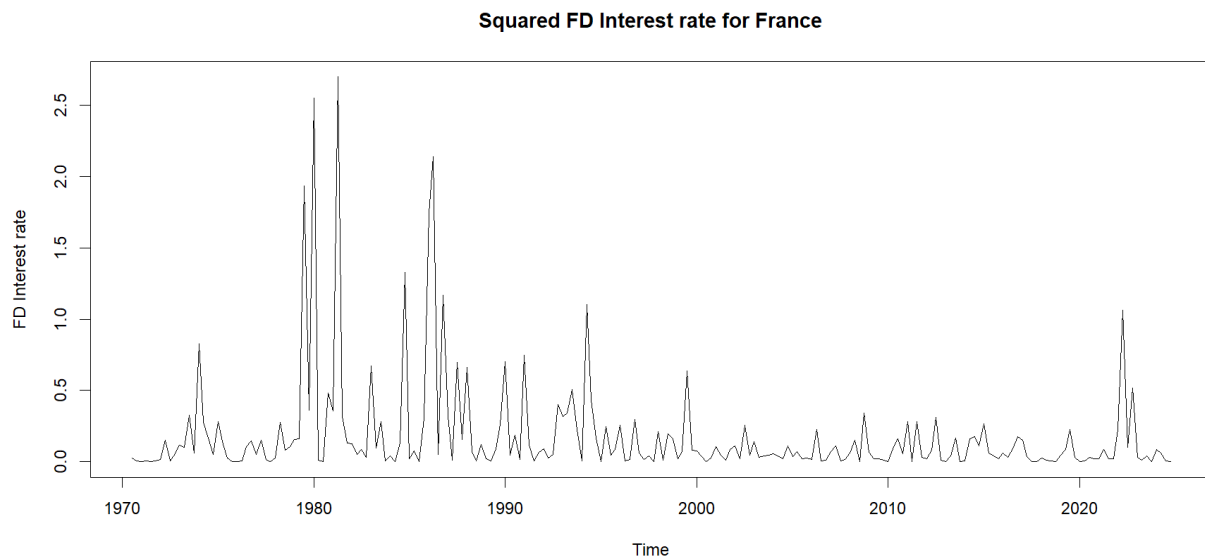
After seasonal difference, cyclical pattern disappears.



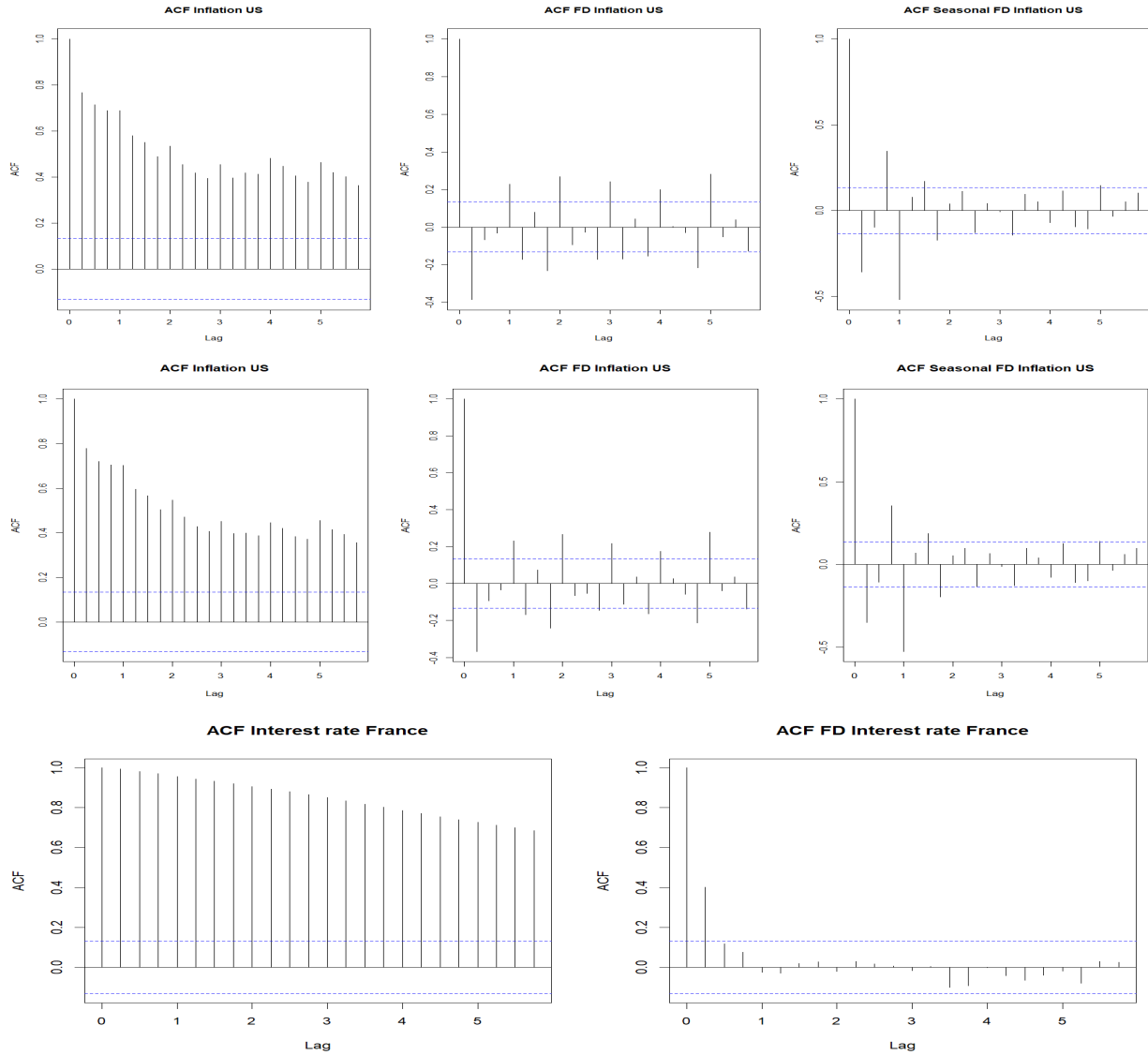
Also, the spectral density appears spread across frequencies.



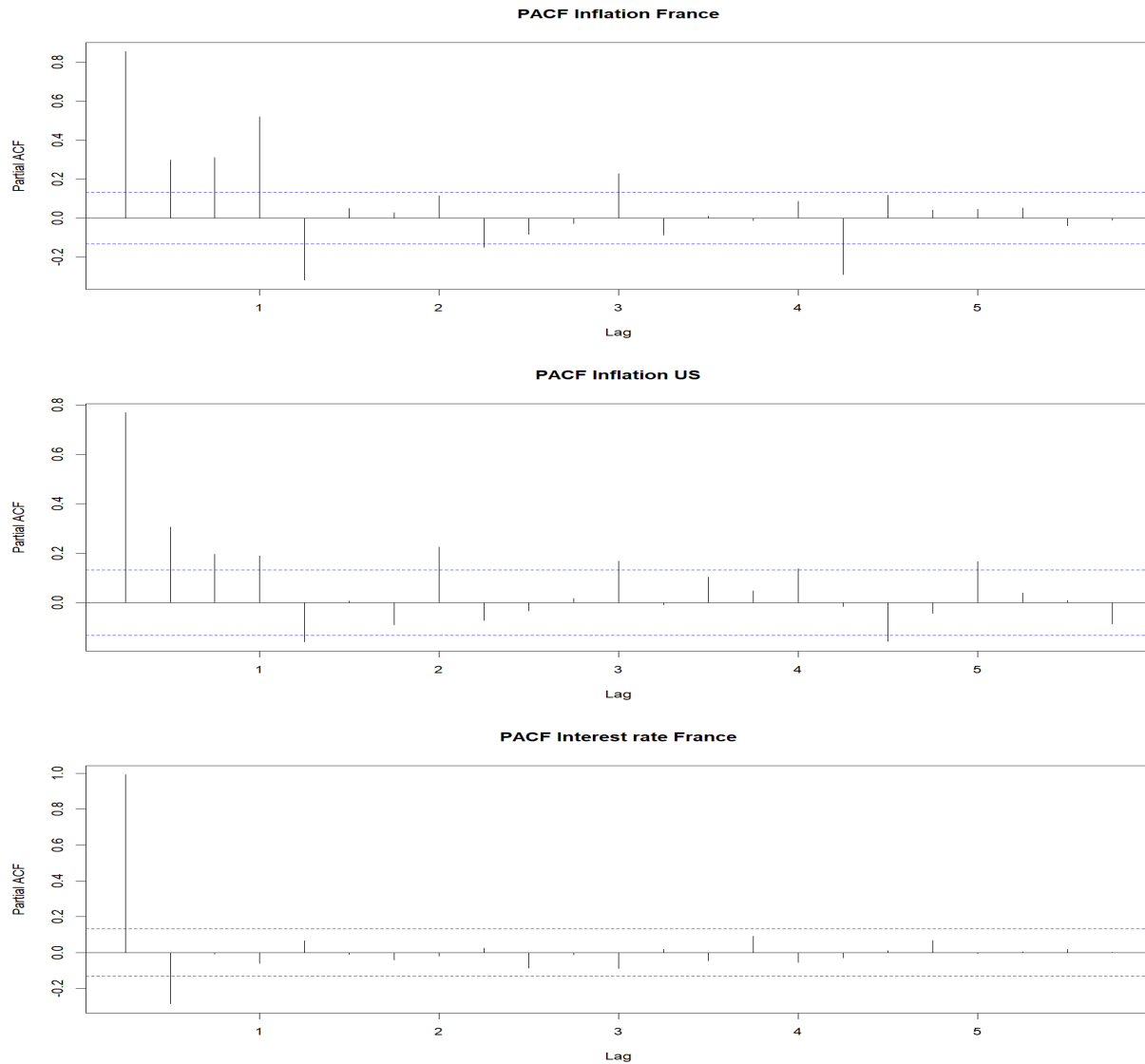
The squared first differences suggest conditional heteroskedasticity in both inflation series, as they show persistent volatility clustering.



Same pattern for french interest rate.



The ACF confirms the previous analysis, as it remains highly persistent until differences.



At level, the PACF suggests that persistence is driven mainly by the first lag.

```
##### QUESTION 2
library(vars) # VAR tools
library(astsa) # time series tools
##### LEVEL GRAPHS
x=ts.intersect(INFFR,INFUS,IR10FR) # align the three series on common
  dates
colnames(x)=c("INFFR","INFUS","IR10FR") # assign variable names
par(mfrow=c(3,1),mar=c(3,4,2,2),oma=c(0,0,2,0)) # 3 stacked plots
tsplot(INFFR,col=couleur,ylab="Inflation",xlab="Time",type="o",main="
  Inflation for France") # France inflation series
tsplot(INFUS,col=couleur,ylab="Inflation",xlab="Time",type="o",main="
  Inflation for United States") # US inflation series
tsplot(IR10FR,col=couleur,ylab="Interest Rates",xlab="Time",type="o",main=
```

```

    "Interest Rates for France") # France interest rate series

##### RAW DATA PERIODOGRAMS
mvspec(x[, "INFFR"], main="Raw Periodogram on raw data for inflation, France
") # raw periodogram for France inflation
mvspec(x[, "INFUS"], main="Raw Periodogram on raw data for inflation, US") #
    raw periodogram for US inflation
mvspec(x[, "IR10FR"], main="Raw Periodogram on raw data for Interest rate,
    France") # raw periodogram for France interest rate

##### FIRST DIFFERENCES
dx=diff(x) # first differences of all variables
plot.ts(dx[, "INFFR"], col=couleur, ylab="FD Inflation", xlab="Time", main="
    First Difference Inflation for France") # first difference of France
    inflation
plot.ts(dx[, "INFUS"], col=couleur, ylab="FD Inflation", xlab="Time", main="
    First Difference Inflation for United States") # first difference of US
    inflation
plot.ts(dx[, "IR10FR"], col=couleur, ylab="FD Interest Rates", xlab="Time",
    main="First Difference Interest Rates for France") # first difference
    of France interest rate

##### SMOOTHED PERIODOGRAMS
graphics.off() # reset plots
par(mfrow=c(3,1)) # prepare stacked plots
mvspec(dx[, "INFFR"], main="Smoothed Periodogram on FD for inflation, France
", kernel=kernel("daniell", 7)) # smoothed FD France inflation
mvspec(dx[, "INFUS"], main="Smoothed Periodogram on FD for inflation, US",
    kernel=kernel("daniell", 7)) # smoothed FD US inflation
mvspec(dx[, "IR10FR"], main="Smoothed Periodogram on FD for Interest rate,
    France", kernel=kernel("daniell", 4)) # smoothed FD France interest rate

##### SEASONAL DIFFERENCES
graphics.off()
par(mfrow=c(3,1))
dINFFRS=diff(dx[, "INFFR"], 4) # seasonal difference of differenced France
    inflation
dINFUSS=diff(dx[, "INFUS"], 4) # seasonal difference of differenced US
    inflation
par(mfrow=c(2,1))
plot.ts(dINFFRS, col=couleur, ylab="FD Inflation", xlab="Time", main="Seasonal
    Difference of France Inflation") # seasonal differenced France
    inflation
plot.ts(dINFUSS, col=couleur, ylab="FD Inflation", xlab="Time", main="Seasonal
    Difference of US Inflation") # seasonal differenced US inflation

##### FIRST DIFFERENCE PERIODOGRAMS
graphics.off() # reset plots
par(mfrow=c(3,1)) # prepare stacked plots
mvspec(dINFFRS, main="Raw Periodogram for inflation (seasonal and no
    seasonal FD), France") # seasonal FD France inflation
mvspec(dINFUSS, main="Raw Periodogram for inflation (seasonal and no
    seasonal FD), US") # seasonal FD US inflation
mvspec(dx[, "IR10FR"], main="Raw Periodogram for Interest rate FD, France")

```

```

# FD France interest rate

##### CONDITIONAL HETEROSCEDASTICITY : SQUARED DIFFERENCES
par(mfrow=c(1,2),mar=c(3,4,2,2),oma=c(0,0,2,0)) # single plot window
plot.ts(dINFRRS^2,col=couleur,ylab="FD Inflation",xlab="Time",main="
Squared seasonal FD Inflation for France") # squared seasonal FD France
inflation
plot.ts(dINFUSS^2,col=couleur,ylab="FD Inflation",xlab="Time",main="
Squared seasonal FD Inflation for United States") # squared seasonal FD
US inflation
plot.ts(dx["IR10FR"]^2,col=couleur,ylab="FD Interest rate",xlab="Time",
main="Squared FD Interest rate for France") # repeated squared FD
France interest rate

```

3 Statistical Analysis

3.1 stationarity:

We follow the Bernhard Pfaff procedure. We consider the 5% critical value.

3.1.1 Inflation in France

```

> ##### UNIT ROOT TESTS : FRANCE INFLATION
> summary(ur.df(INFFR,lags=8,type="trend",selectlags="AIC")) # level with trend
Test regression trend
Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-6.773 -0.736 -0.170  0.640  5.136
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.074890    0.465864   2.307 0.022067 *
z.lag.1      -0.120502    0.046339  -2.600 0.010010 *
tt           -0.005876    0.002693  -2.181 0.030321 *
z.diff.lag1 -0.331606    0.077947  -4.254 3.23e-05 ***
z.diff.lag2 -0.316972    0.081304  -3.899 0.000132 ***
z.diff.lag3 -0.252528    0.084125  -3.002 0.003028 **
z.diff.lag4  0.259668    0.085460   3.038 0.002697 **
z.diff.lag5 -0.004300    0.087334  -0.049 0.960780
z.diff.lag6  0.009771    0.083906   0.116 0.907416
z.diff.lag7 -0.008080    0.078771  -0.103 0.918405
z.diff.lag8  0.209697    0.071518   2.932 0.003761 **

```

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.415 on 199 degrees of freedom
Multiple R-squared:  0.548, Adjusted R-squared:  0.5253
F-statistic: 24.12 on 10 and 199 DF,  p-value: < 2.2e-16

```

Value of test-statistic is: -2.6004 2.3181 3.3944

Critical values for test statistics:

```

      1pct  5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2  6.22  4.75  4.07
phi3  8.43  6.49  5.47

```

The τ_3 statistic (-2.600) is above the critical value (-3.43), so we fail to reject the null of a unit root. The ϕ_3 statistic (3.394) is below the critical value (6.49), so we fail to reject the joint null hypothesis of a unit root and no deterministic trend. We retain a model with a constant but no trend.

```

> summary(ur.df(INFFR,lags=8,type="drift",selectlags="AIC")) # level with drift
Test regression drift
Call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-7.0742 -0.7797 -0.1427  0.6312  5.2827
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.10699    0.14335   0.746  0.45630
z.lag.1      -0.03950    0.02798  -1.412  0.15963
z.diff.lag1  -0.39507    0.07299  -5.413 1.77e-07 ***
z.diff.lag2  -0.37185    0.07804  -4.765 3.62e-06 ***
z.diff.lag3  -0.29798    0.08227  -3.622 0.00037 ***
z.diff.lag4   0.22336    0.08461   2.640 0.00895 **
z.diff.lag5  -0.05026    0.08555  -0.588 0.55749
z.diff.lag6  -0.02911    0.08276  -0.352 0.72536
z.diff.lag7  -0.03652    0.07841  -0.466 0.64195
z.diff.lag8   0.19198    0.07172   2.677 0.00805 **

```

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.428 on 200 degrees of freedom
Multiple R-squared:  0.5372, Adjusted R-squared:  0.5163
F-statistic: 25.79 on 9 and 200 DF,  p-value: < 2.2e-16

```

Value of test-statistic is: -1.4116 1.0775

Critical values for test statistics:

```

1pct  5pct 10pct
tau2 -3.46 -2.88 -2.57
phi1  6.52  4.63  3.81

```

The τ_{u_2} statistic (-1.412) is above the critical value (-2.88), we fail to reject the null of unit root. The ϕ_{i_1} statistic (1.078) is below the critical value of 4.63. we fail to reject this joint null and retain a model with no trend and no constant.

```

> summary(ur.df(INFFR,lags=8,type="none",selectlags="AIC")) # level without deterministic term
Test regression none
Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-7.1502 -0.6893 -0.0644  0.6794  5.2717
Coefficients:
Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.02435     0.01924  -1.265 0.207169
z.diff.lag1 -0.40782     0.07089  -5.753 3.23e-08 ***
z.diff.lag2 -0.38305     0.07650  -5.007 1.20e-06 ***
z.diff.lag3 -0.30785     0.08111  -3.795 0.000195 ***
z.diff.lag4  0.21488     0.08375   2.566 0.011024 *
z.diff.lag5 -0.05969     0.08452  -0.706 0.480844
z.diff.lag6 -0.03642     0.08209  -0.444 0.657792
z.diff.lag7 -0.04219     0.07796  -0.541 0.588931
z.diff.lag8  0.18820     0.07146   2.634 0.009106 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.427 on 201 degrees of freedom
Multiple R-squared:  0.536, Adjusted R-squared:  0.5152
F-statistic: 25.8 on 9 and 201 DF,  p-value: < 2.2e-16

```

Value of test-statistic is: -1.2655

Critical values for test statistics:

```

1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

τ_{u_1} test statistic (-1.266), is greater than the critical value (-1.95), we fail to reject the null hypothesis of a unit root. We take first differences.

```

> summary(ur.df(dINFFR,lags=8,type="trend",selectlags="AIC")) # difference with trend
Test regression trend

```

```

Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-7.3809 -0.6748 -0.0801  0.6407  5.1405
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0660281  0.2095436  -0.315    0.7530
z.lag.1      -1.9486187  0.3644364  -5.347 2.44e-07 ***
tt           0.0001152  0.0016354   0.070   0.9439
z.diff.lag1  0.5192937  0.3388667   1.532   0.1270
z.diff.lag2  0.1250476  0.3077538   0.406   0.6849
z.diff.lag3 -0.1956425  0.2719923  -0.719   0.4728
z.diff.lag4  0.0065753  0.2347118   0.028   0.9777
z.diff.lag5 -0.0696288  0.1798264  -0.387   0.6990
z.diff.lag6 -0.1280117  0.1253610  -1.021   0.3084
z.diff.lag7 -0.1843029  0.0713377  -2.584   0.0105 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.426 on 199 degrees of freedom
Multiple R-squared:  0.8278, Adjusted R-squared:  0.82
F-statistic: 106.3 on 9 and 199 DF,  p-value: < 2.2e-16

```

Value of test-statistic is: -5.3469 9.5602 14.2951

Critical values for test statistics:

```

1pct  5pct 10pct
tau3  -3.99 -3.43 -3.13
phi2   6.22  4.75  4.07
phi3   8.43  6.49  5.47

```

The τ_3 statistic (-5.347) is below the critical value (-3.43), so we reject the unit root null. As neither the trend nor the intercept is significant, We retain the series without trend or drift.

```

> summary(ur.df(dINFFR,lags=8,type="none",selectlags="AIC")) # difference without deterministic
Test regression none
Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-7.4417 -0.7233 -0.1249  0.5845  5.0696
Coefficients:
Estimate Std. Error t value Pr(>|t|)
z.lag.1      -1.939734  0.362515  -5.351 2.38e-07 ***
z.diff.lag1  0.511294  0.337100   1.517  0.13090
z.diff.lag2  0.118160  0.306178   0.386  0.69996

```

```

z.diff.lag3 -0.200982    0.270650   -0.743   0.45860
z.diff.lag4  0.003009    0.233611    0.013   0.98974
z.diff.lag5 -0.071806    0.179008   -0.401   0.68875
z.diff.lag6 -0.129189    0.124808   -1.035   0.30187
z.diff.lag7 -0.184786    0.071028   -2.602   0.00997 **

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.42 on 201 degrees of freedom

Multiple R-squared: 0.8276, Adjusted R-squared: 0.8207

F-statistic: 120.6 on 8 and 201 DF, p-value: < 2.2e-16

Value of test-statistic is: -5.3508

Critical values for test statistics:

```

      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

Same, the first-differenced inflation series for France is stationary around zero.

3.1.2 Inflation in the US and Interest rate in France

Results for US inflation and the French interest rate (reported in appendix) follow the same pattern and are summarized below. Both series are stationary around zero after first differencing.

Transformation	Specification	Rejection of H_0	Conclusion / Next Step
Level	Trend	No (τ_{u3}, ϕ_{i3})	Retain constant without trend
Level	Drift	No (τ_{u2}, ϕ_{i1})	Retain without trend or constant
Level	None	No (τ_{u1})	Take first differences
1st Difference	Trend	Yes (τ_{u3})	Intercept/Trend not significant, re-test
1st Difference	None	Yes (τ_{u1})	Series is stationary around zero

3.2 Breaks

```

> sctest(INFFR~t,type="Chow",point=c(1985,1)) # Chow test in level with trend
Chow test
data:  INFFR ~ t
F = 85.486, p-value < 2.2e-16

```

```
> sctest(dINFFR~1,type="Chow",point=c(1985,1)) # Chow test in first difference
Chow test
data:  dINFFR ~ 1
F = 0.021323, p-value = 0.884
```

A 1985 Chow test with a trend, on French inflation shows a structural break in levels, but not in first differences (without trend).

```
> sctest(INFUS~t,type="Chow",point=c(1983,1)) # Chow test in level with trend
Chow test
data:  INFUS ~ t
F = 39.367, p-value = 2.665e-15
```

```
> sctest(dINFUS~1,type="Chow",point=c(1983,1)) # Chow test in first difference
Chow test
data:  dINFUS ~ 1
F = 0.16046, p-value = 0.6891
```

The same pattern for US inflation in 1983.

3.3 Seasonality

Table 1: Quarterly means before and after differencing

Series	Q1	Q2	Q3	Q4
<i>Panel A: Levels</i>				
France inflation	3.157	5.125	3.369	3.109
US inflation	3.568	4.256	3.556	3.942
France interest rate	6.454	6.505	6.489	6.468
<i>Panel B: First differences</i>				
France inflation	-0.047	1.973	-1.756	-0.260
US inflation	-0.388	0.638	-0.700	0.386
France interest rate	-0.078	0.012	-0.015	-0.021
<i>Panel C: Seasonal differences</i>				
France inflation	0.029	0.054	-0.045	-0.077
US inflation	0.128	-0.044	0.009	-0.032

A descriptive comparison of quarterly means suggests non-random differences across quarters, with French inflation being up to 2% higher on average in Q2. After seasonal differencing, this pattern disappears.

```
##### QUESTION 3
##### STATIONARITY
library(urca) # unit root tests
library(strucchange) # Chow break tests for univariate series
dx=diff(x) # first differences again for safety
dINFFR=dx[, "INFFR"] # differenced France inflation
dINFUS=dx[, "INFUS"] # differenced US inflation
dIR10FR=dx[, "IR10FR"] # differenced France interest rate

##### UNIT ROOT TESTS : FRANCE INFLATION
summary(ur.df(INFRR, lags=8, type="trend", selectlags="AIC")) # level with
trend
summary(ur.df(INFRR, lags=8, type="drift", selectlags="AIC")) # level with
drift
summary(ur.df(INFRR, lags=8, type="none", selectlags="AIC")) # level without
deterministic term
summary(ur.df(dINFRR, lags=8, type="trend", selectlags="AIC")) # difference
with trend
```

```

summary(ur.df(dINFFR,lags=8,type="none",selectlags="AIC")) # difference
without deterministic term

##### UNIT ROOT TESTS : US INFLATION
summary(ur.df(INFUS,lags=8,type="trend",selectlags="AIC")) # level with
trend
summary(ur.df(INFUS,lags=8,type="drift",selectlags="AIC")) # level with
drift
summary(ur.df(INFUS,lags=8,type="none",selectlags="AIC")) # level without
deterministic term
summary(ur.df(dINFUS,lags=8,type="trend",selectlags="AIC")) # difference
with trend
summary(ur.df(dINFUS,lags=8,type="none",selectlags="AIC")) # difference
without deterministic term

##### UNIT ROOT TESTS : FRANCE INTEREST RATE
summary(ur.df(IR10FR,lags=8,type="trend",selectlags="AIC")) # level with
trend
summary(ur.df(IR10FR,lags=8,type="drift",selectlags="AIC")) # level with
drift
summary(ur.df(IR10FR,lags=8,type="none",selectlags="AIC")) # level without
deterministic term
summary(ur.df(dIR10FR,lags=8,type="trend",selectlags="AIC")) # difference
with trend
summary(ur.df(dIR10FR,lags=8,type="none",selectlags="AIC")) # difference
without deterministic term

##### BREAK TESTS
##### CHOW TEST : FRANCE
t=1:length(INFFR) # time trend for France inflation
sctest(INFFR~t,type="Chow",point=c(1985,1)) # Chow test in level with
trend
sctest(dINFFR~1,type="Chow",point=c(1985,1)) # Chow test in first
difference

##### CHOW TEST : US
t=1:length(INFUS) # time trend for US inflation
sctest(INFUS~t,type="Chow",point=c(1983,1)) # Chow test in level with
trend
sctest(dINFUS~1,type="Chow",point=c(1983,1)) # Chow test in first
difference

##### SEASONALITY CHECK
tapply(INFFR,cycle(INFFR),mean,na.rm=TRUE) # average inflation by quarter
for France
tapply(INFUS,cycle(INFUS),mean,na.rm=TRUE) # average inflation by quarter
for US
tapply(IR10FR,cycle(IR10FR),mean,na.rm=TRUE) # average interest rate by
quarter for France

tapply(dINFFR,cycle(dINFFR),mean,na.rm=TRUE) # average inflation by
quarter for France
tapply(dINFUS,cycle(dINFUS),mean,na.rm=TRUE) # average inflation by
quarter for US

```

```
tapply(dIR10FR, cycle(dIR10FR), mean, na.rm=TRUE) # average interest rate by
quarter for France

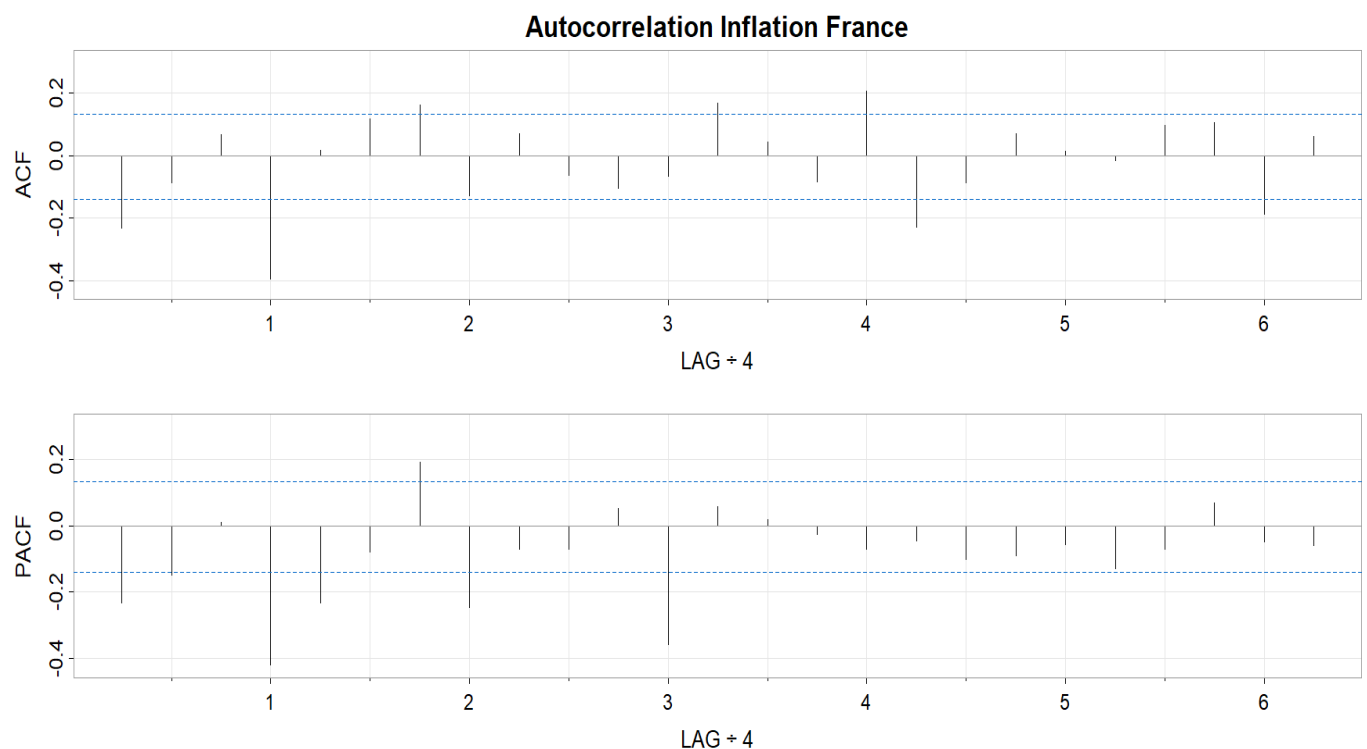
tapply(dINFFRS, cycle(dINFFRS), mean, na.rm=TRUE) # average inflation by
quarter for France

tapply(dINFUSS, cycle(dINFUSS), mean, na.rm=TRUE) # average inflation by
quarter for US
```

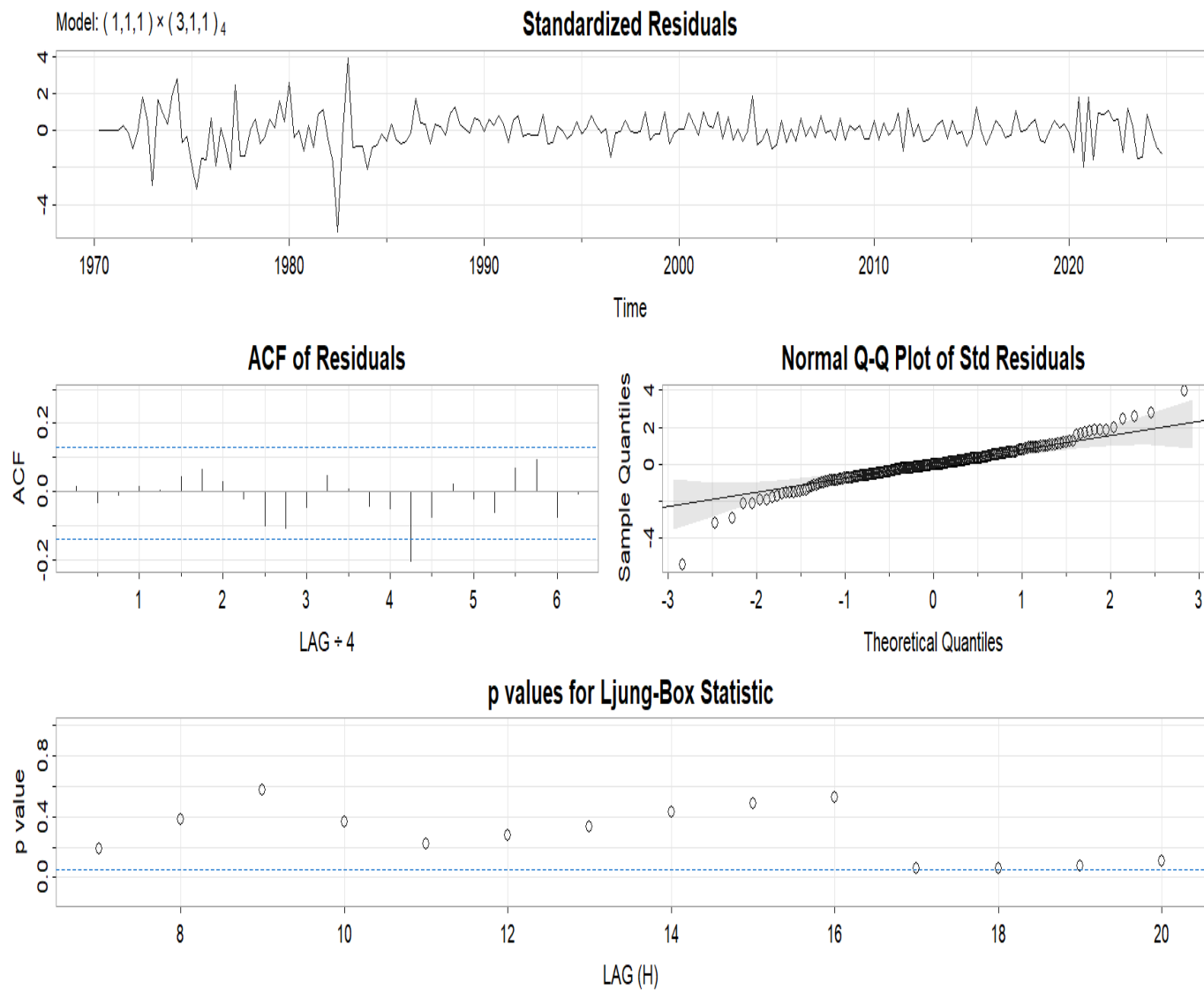
4 Univariate ARIMA models

Previous results suggested one seasonal and non seasonal differences in both inflation.

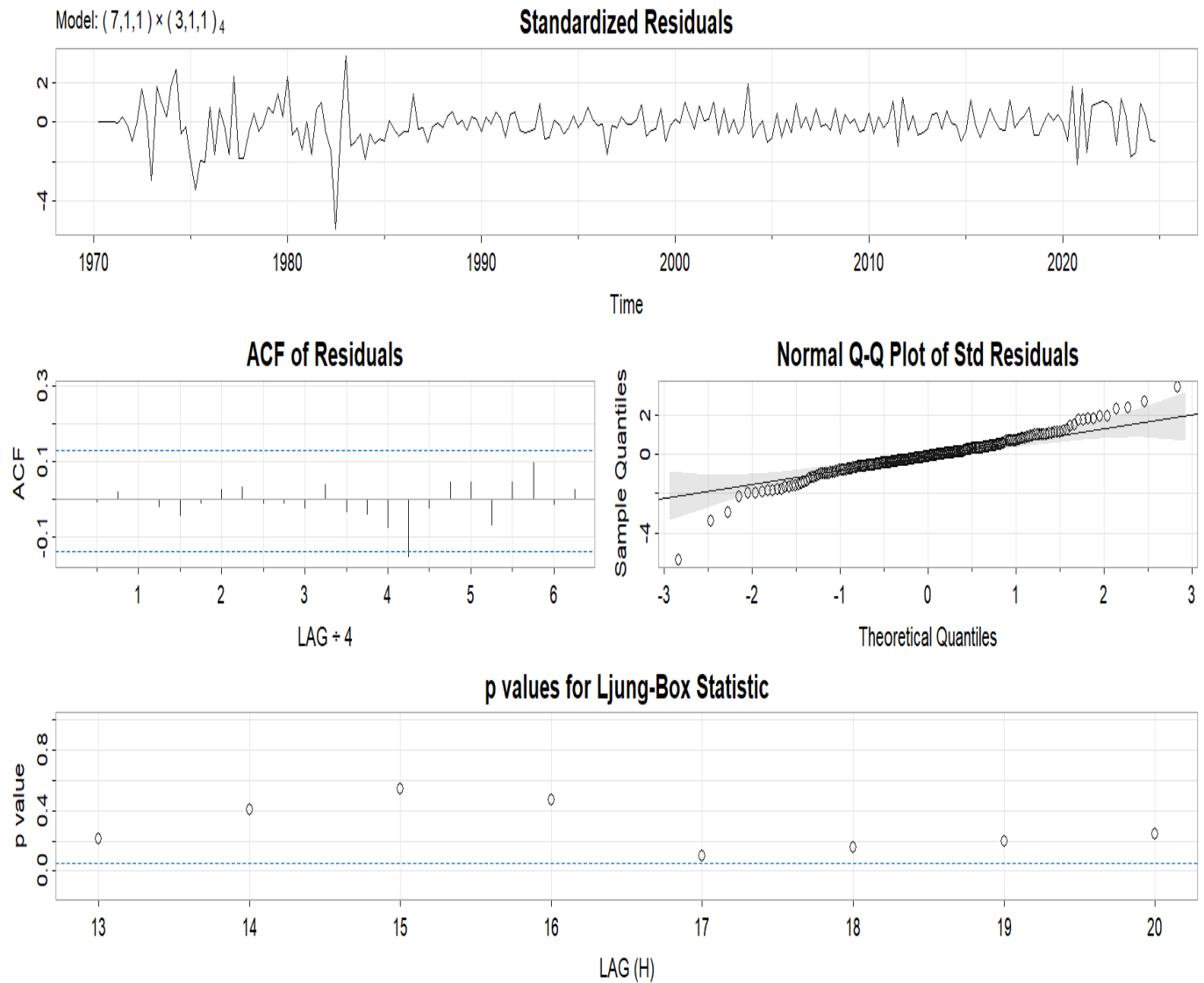
4.1 Domestic inflation



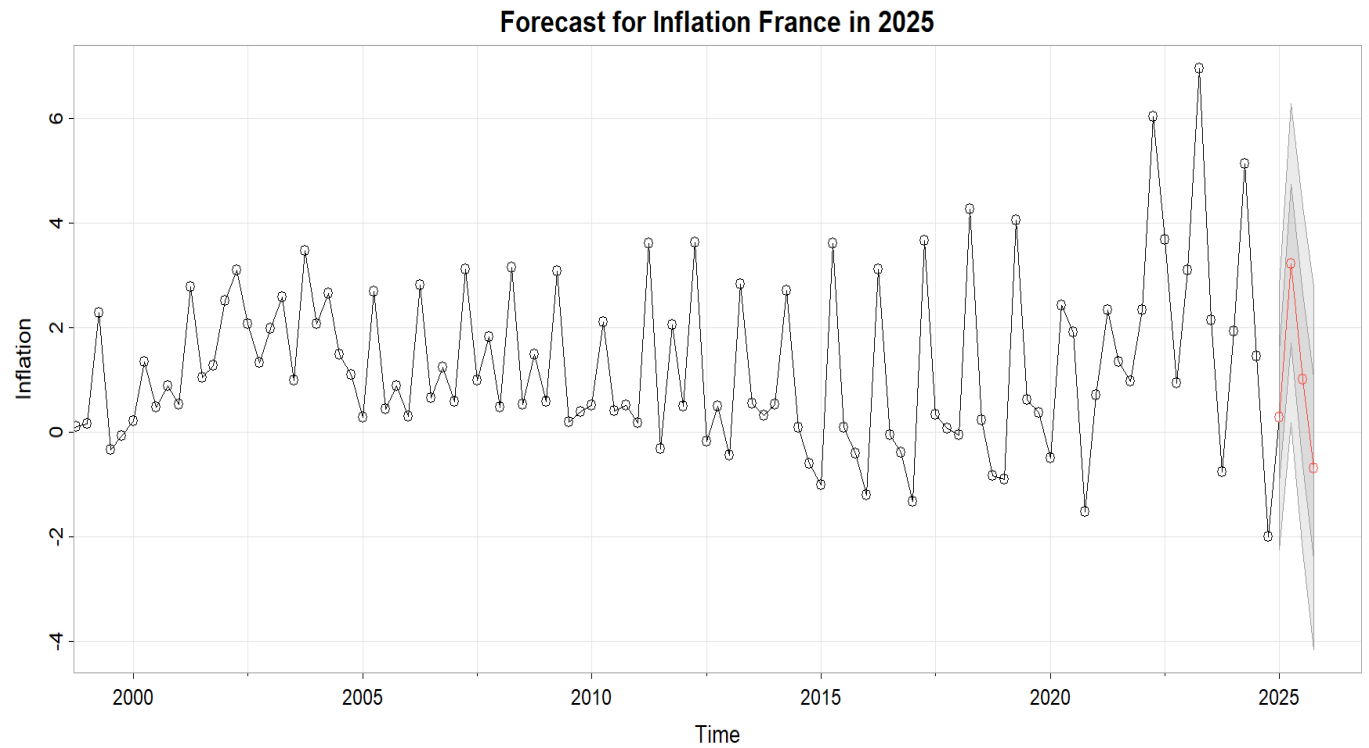
Not clear, but one strong non-seasonal spike in both the ACF and PACF suggests an non-seasonal ARMA(1,1). One strong seasonal spike in the ACF and three in the PACF suggest a seasonal ARMA(3,1).



Because of one weak residual spike, we select the non-seasonal AR term using the AIC.



AIC suggests a non-seasonal AR(7), and the figure above shows that this specification performs better in terms of the residual ACF, although the Ljung-Box test is not satisfactory. Thus, we estimate a non-seasonal ARMA(7,1) with a seasonal ARMA(3,1).



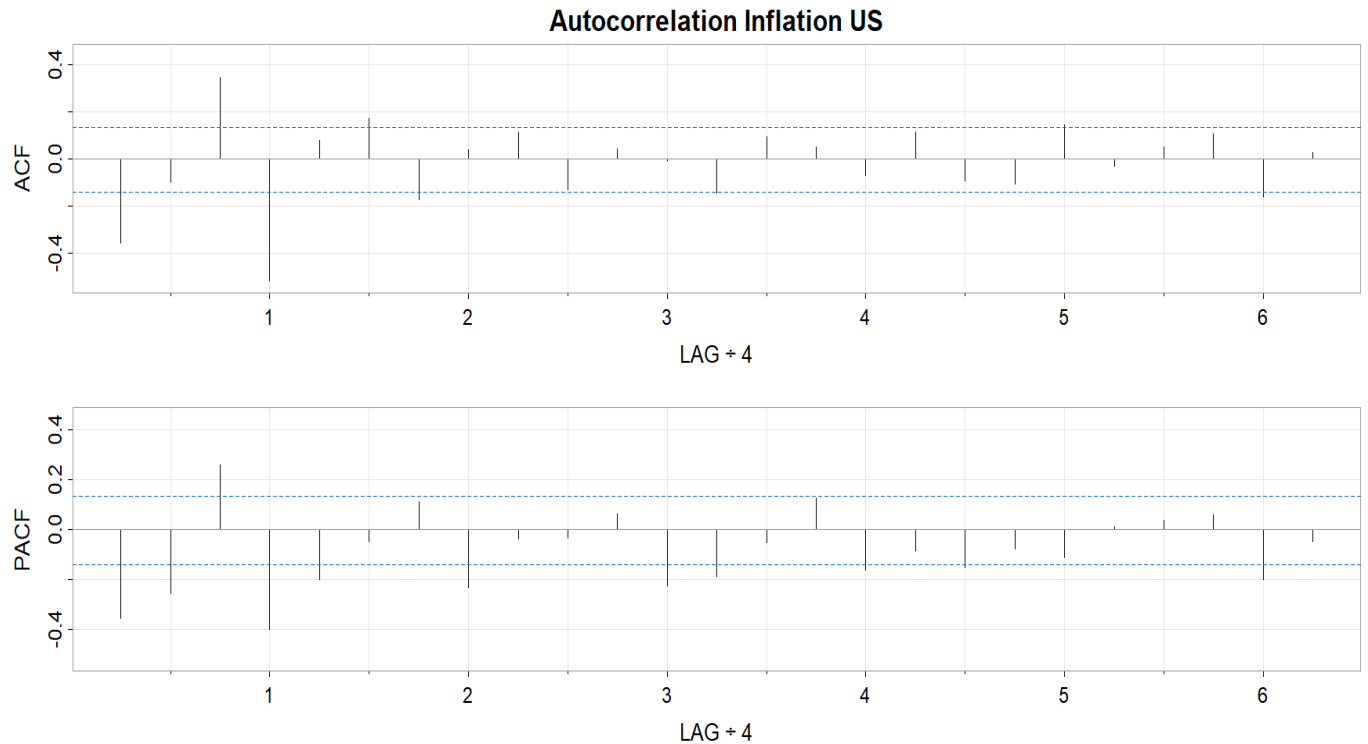
Inflation is expected to decline in a cyclical pattern following the post-Covid downward trend.

Table 2: Estimated SARIMA $(7, 1, 1) \times (3, 1, 1)_4$ model

Parameter	Estimate	SE	t-value	p-value
ar1	0.6718	0.0710	9.4634	0.0000
ar2	0.0421	0.0834	0.5043	0.6146
ar3	0.0418	0.0765	0.5458	0.5858
ar4	-0.5978	0.0765	-7.8129	0.0000
ar5	0.5152	0.0892	5.7777	0.0000
ar6	0.0887	0.0838	1.0587	0.2910
ar7	0.0332	0.0727	0.4558	0.6490
ma1	-1.0000	0.0239	-41.8700	0.0000
sar1	0.9178	0.1269	7.2344	0.0000
sar2	-0.5692	0.1053	-5.4059	0.0000
sar3	0.4858	0.1061	4.5806	0.0000
sma1	-0.9482	0.0869	-10.9179	0.0000

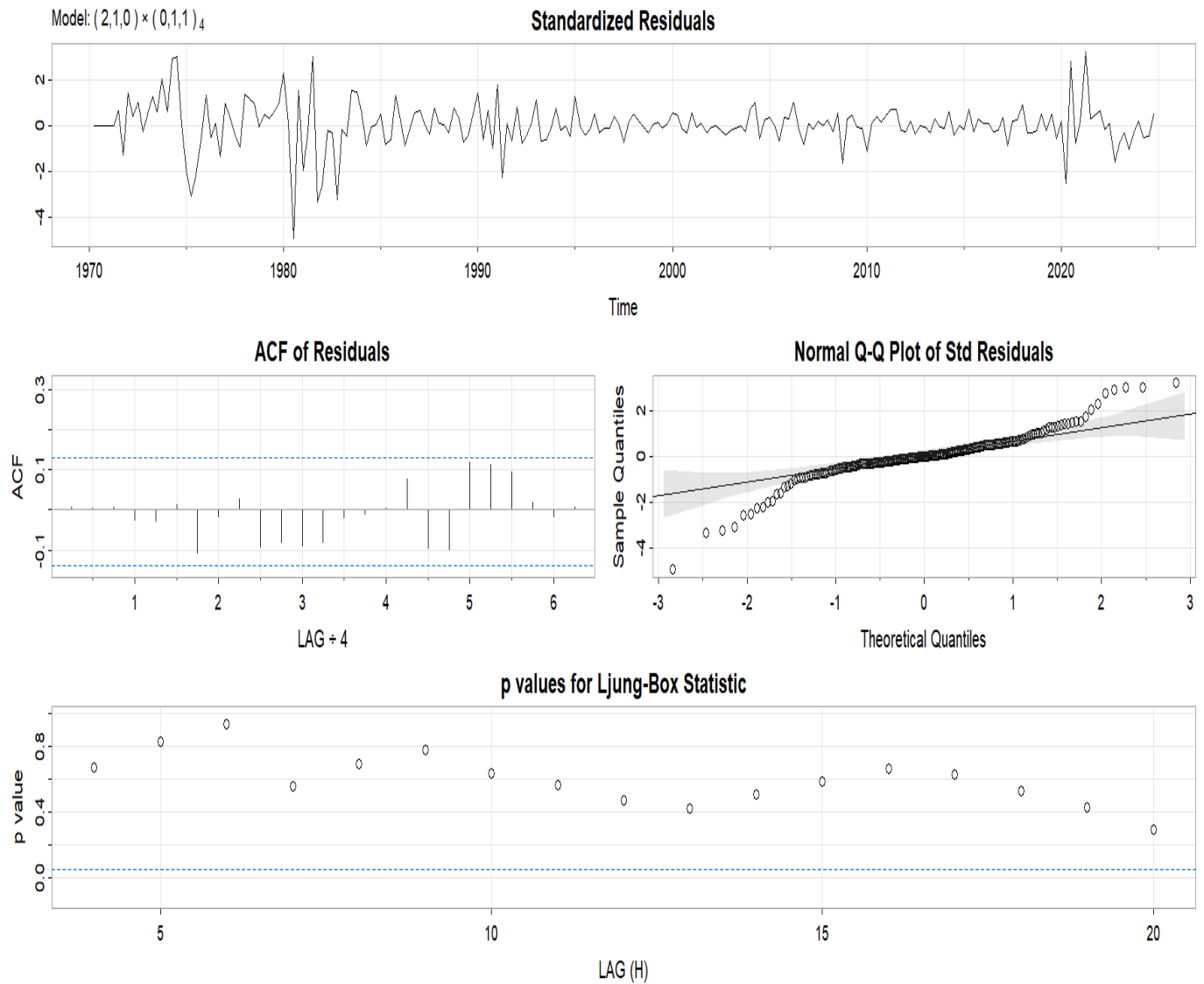
$\hat{\sigma}^2 = 1.582076$, AIC = 3.458254, AICc = 3.465508, BIC = 3.662729.

4.2 Foreign inflation : US

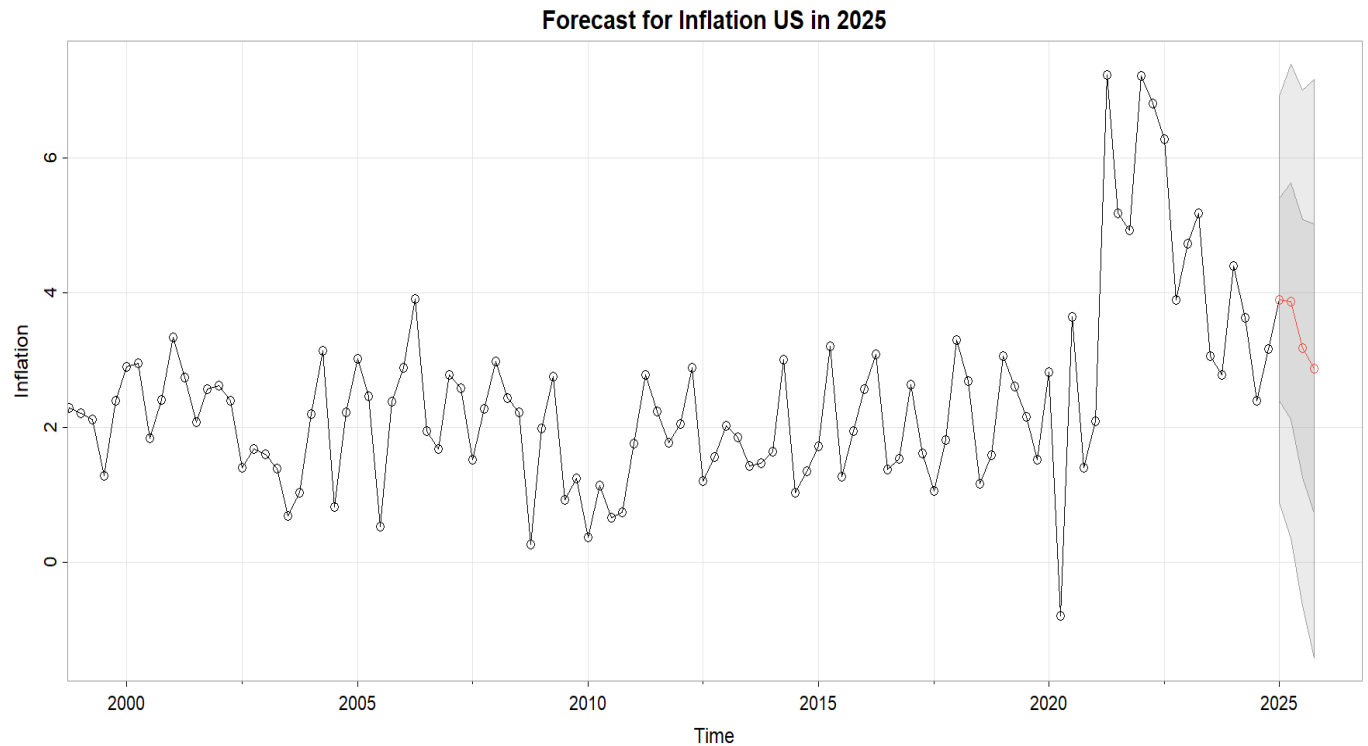


Non-seasonal : One spike in ACF and two or three spikes in the PACF. Since the MA term was not significant in estimation, we estimate a non-seasonal AR(2).

Seasonal : One strong seasonal spike in the ACF, with the PACF trailing off, suggests a seasonal MA(1).



Both the residual ACF and the Ljung-Box test are satisfactory.



Inflation is expected to decline in 2025, continuing the post-Covid downward trend. Q2 is not forecast to be the highest quarter, suggesting that recent macroeconomic pattern can dominate the usual seasonal pattern. The two negative AR components imply that the last two increases will slow the Q2 inflation.

Table 3: Estimated SARIMA $(2, 1, 0) \times (0, 1, 1)_4$ model

Coefficient	Estimate	SE	t-value	p-value
<i>ar1</i>	-0.4014	0.0658	-6.0965	0.0000
<i>ar2</i>	-0.2597	0.0665	-3.9037	0.0001
<i>sma1</i>	-0.8549	0.0337	-25.3439	0.0000

$$\hat{\sigma}^2 = 2.277483$$

$$\text{AIC} = 3.724094, \text{AICc} = 3.724628, \text{BIC} = 3.78701$$

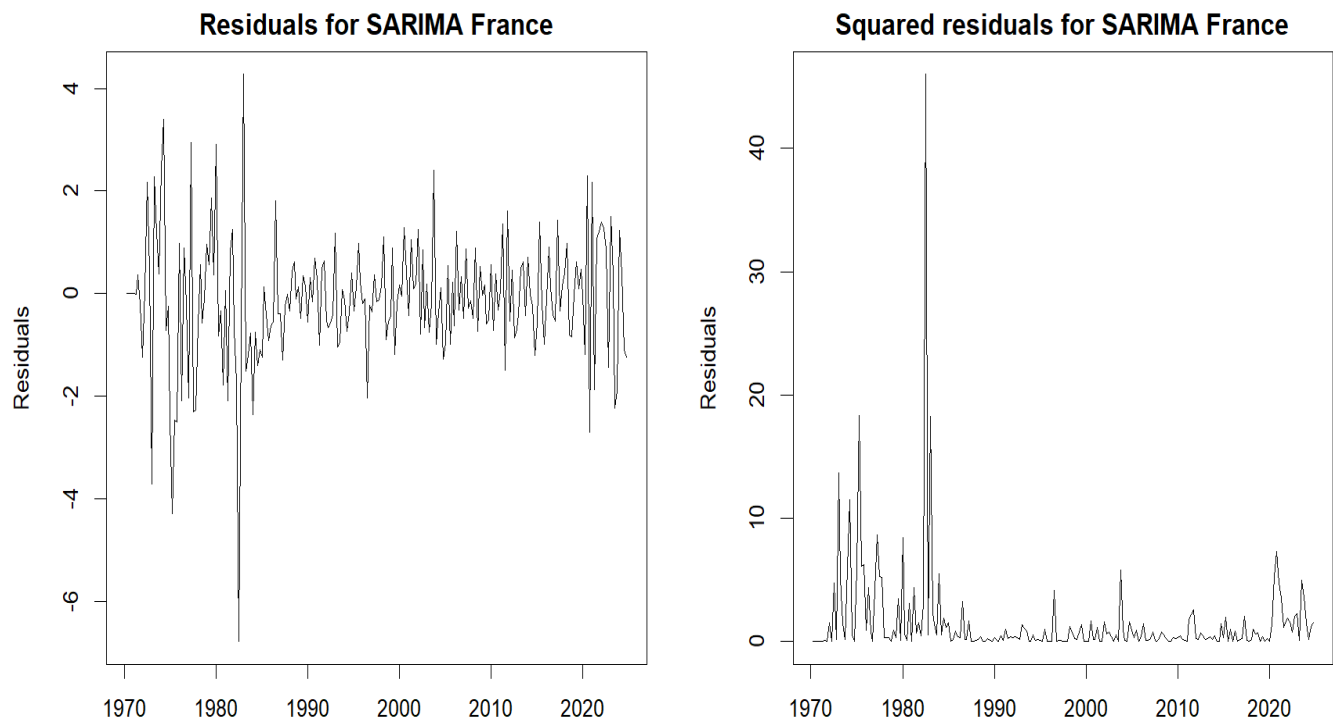
```
##### QUESTION 4
library(astsa) # time series estimation tools
```

```
##### ACF AND PACF : HOME INFLATION (FRANCE)
graphics.off() # close all open graphs
acf2(dINFRRS,main="Autocorrelation Inflation France") # ACF and PACF of
    seasonal differenced France inflation
sarima(INFRR,1,1,1, 3,1,1, 4) # candidate SARIMA model for France
aic_stats=numeric(13) # vector to store AIC values
bic_stats=numeric(13) # vector to store BIC values
for(p in 0:12) { # loop over non-seasonal AR order
    m=sarima(INFRR,p,1,1,3,1,1,4) # estimate SARIMA
    aic_stats[p+1]=m$ICs[1] # save AIC
    bic_stats[p+1]=m$ICs[3] # save BIC
}
which.min(aic_stats)-1 # best p by AIC
which.min(bic_stats)-1 # best p by BIC
sarima(INFRR,7,1,1,3,1,1,4) # selected France model
fcINFRR=sarima.for(INFRR,4, 7,1,1, 3,1,1,4,main="Forecast for Inflation
    France in 2025",ylab="Inflation",xlab="Time") # 4-quarter forecast for
    France inflation

##### ACF AND PACF : FOREIGN INFLATION (US)
acf2(dINFUSS,main="Autocorrelation Inflation US") # ACF and PACF of
    seasonal differenced US inflation
sarima(INFUS,2,1,0, 0,1,1,4) # candidate SARIMA model for US aic_stats=
    numeric(13) # vector to store AIC values
fcINFUS=sarima.for(INFUS,4,2,1,0,0,1,1,4,main="Forecast for Inflation US
    in 2025",ylab="Inflation",xlab="Time") # 4-quarter forecast for US
    inflation
```

5 Conditionally heteroskedastic of residuals

5.1 Domestic inflation : France



Both the residuals and squared residuals show volatility clustering, as periods of low and high variance tend to cluster over time.

```
1 Box.test(ResFR,lag=8) # residual autocorrelation test
```

```
Box-Pierce test  
data: ResFR  
X-squared = 0.71147, df = 8, p-value = 0.9995
```

The portmanteau test fails to reject the null, indicating no autocorrelation in the residuals.

```
1 ArchTest(ResFR,8) # Engle ARCH test
```

```
ARCH LM-test; Null hypothesis: no ARCH effects  
data: ResFR  
Chi-squared = 20.772, df = 8, p-value = 0.007778
```

```

1 ResFR2=ResFR^2 # squared residuals
2 Box.test(ResFR2,lag=8) # autocorrelation test on squared residuals
3

```

Box-Pierce test

data: ResFR2

X-squared = 31.171, df = 8, p-value = 0.000131

Both the Portmanteau and Engle tests confirm the presence of ARCH effects in the squared residuals. We estimate the simplest model suggested by the BIC, a GARCH(2,1) model, implying that current volatility depends on the two previous squared shocks and one lagged conditional variance.

```

1 fitFR=garchFit(ResFR~garch(2,1),data=ResFR,include.mean=FALSE,trace=FALSE)
  # estimate Garch for France residuals

```

GARCH Modelling

Coefficient(s):

omega	alpha1	alpha2	beta1
0.10523926	0.26252263	0.00000001	0.69560101

Std. Errors:

based on Hessian

Error Analysis:

	Estimate	Std. Error	t value	Pr(> t)
omega	0.10523926	0.11658126	0.903	0.36668
alpha1	0.26252263	0.12382584	2.120	0.03400 *
alpha2	0.00000001	0.18784571	0.000	1.00000
beta1	0.69560101	0.25026222	2.779	0.00544 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:

-329.0367 normalized: -1.50245

Standardised Residuals Tests:

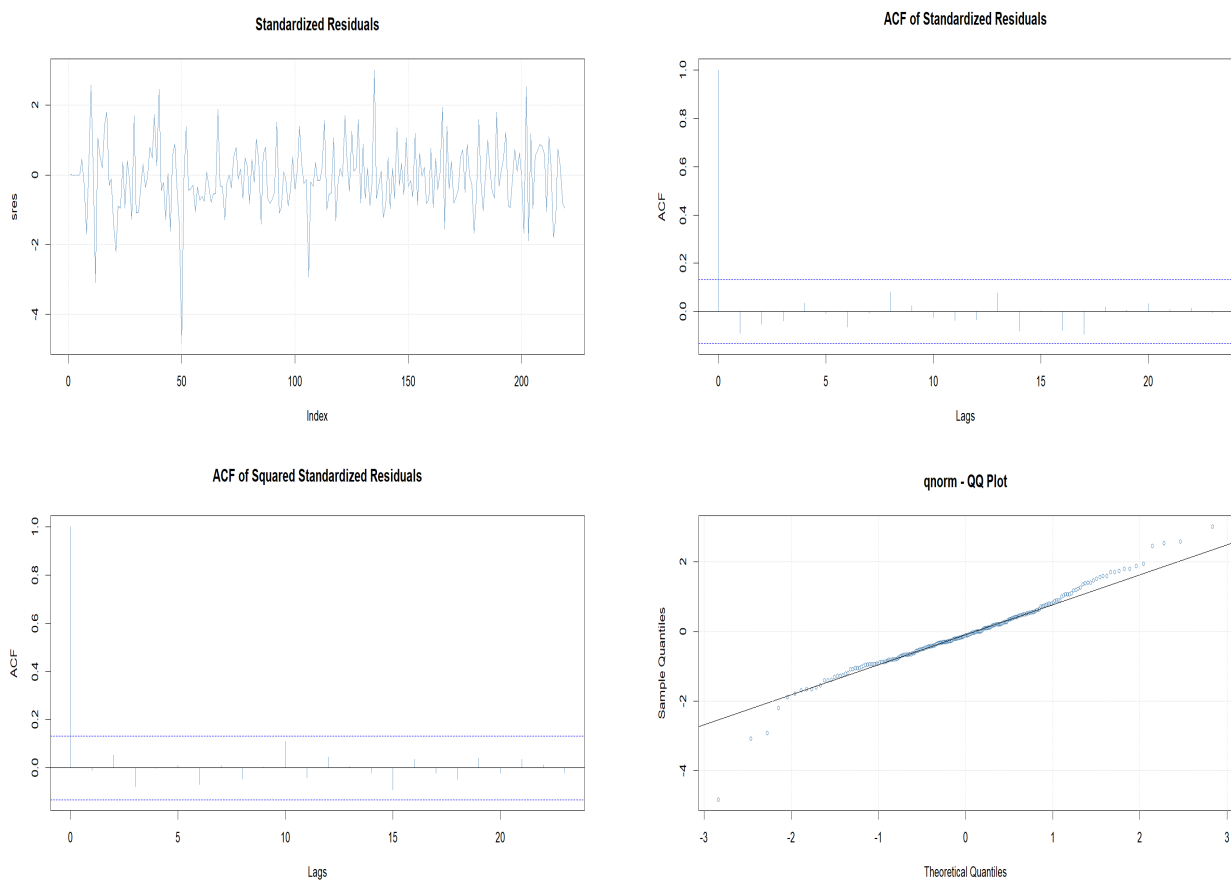
Statistic p-Value

Jarque-Bera Test	R	Chi^2	59.5176791	1.191269e-13
Shapiro-Wilk Test	R	W	0.9683598	8.163248e-05
Ljung-Box Test	R	Q(10)	5.7571782	8.352347e-01
Ljung-Box Test	R	Q(15)	9.1796310	8.679392e-01
Ljung-Box Test	R	Q(20)	13.0255368	8.762857e-01
Ljung-Box Test	R^2	Q(10)	6.3588289	7.842691e-01

Ljung-Box Test	R^2	$Q(15)$	9.2533789	$8.639006e-01$
Ljung-Box Test	R^2	$Q(20)$	10.7436799	$9.526007e-01$
LM Arch Test	R	TR^2	6.5081281	$8.883365e-01$

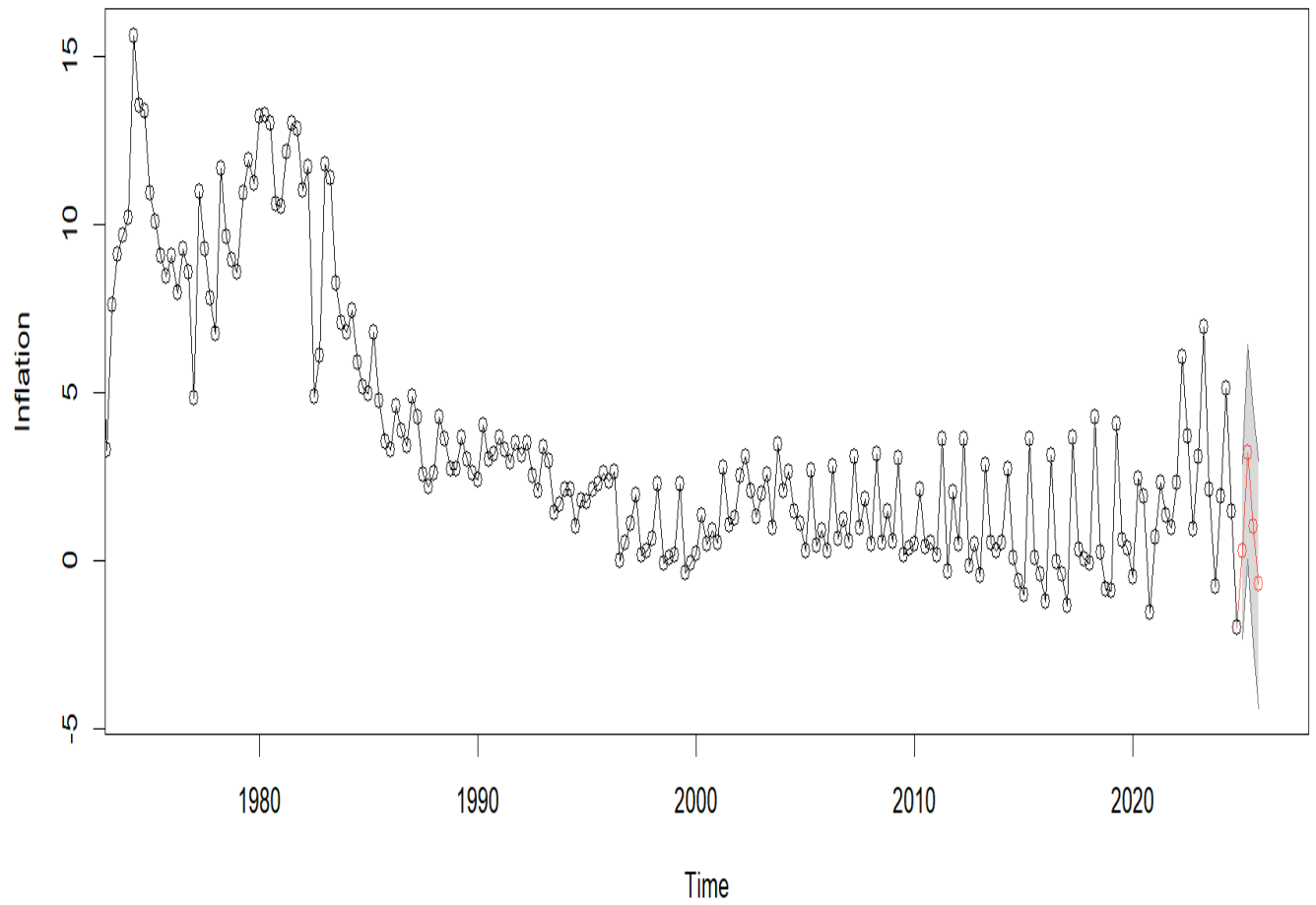
The Ljung-Box tests do not reject the null for the standardised residuals, and the LM ARCH test does not reject the null for their squares. The model captures conditional heteroskedasticity adequately. The coefficient on β_{a_1} (0.696) is relatively large, highlighting the importance of the lagged conditional variance.

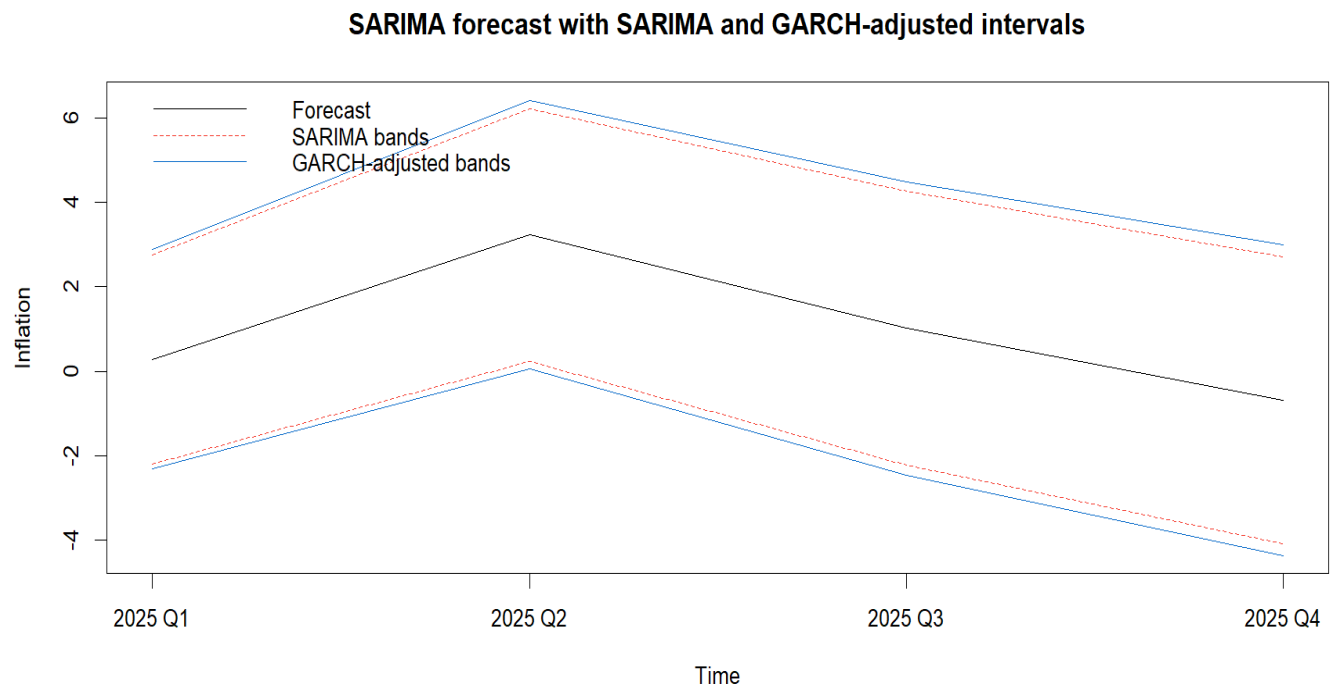
Residuals diagnostic plots



Residual diagnostics show no visible clustering, and the ACF indicates no autocorrelation (the large spike is at lag 0).

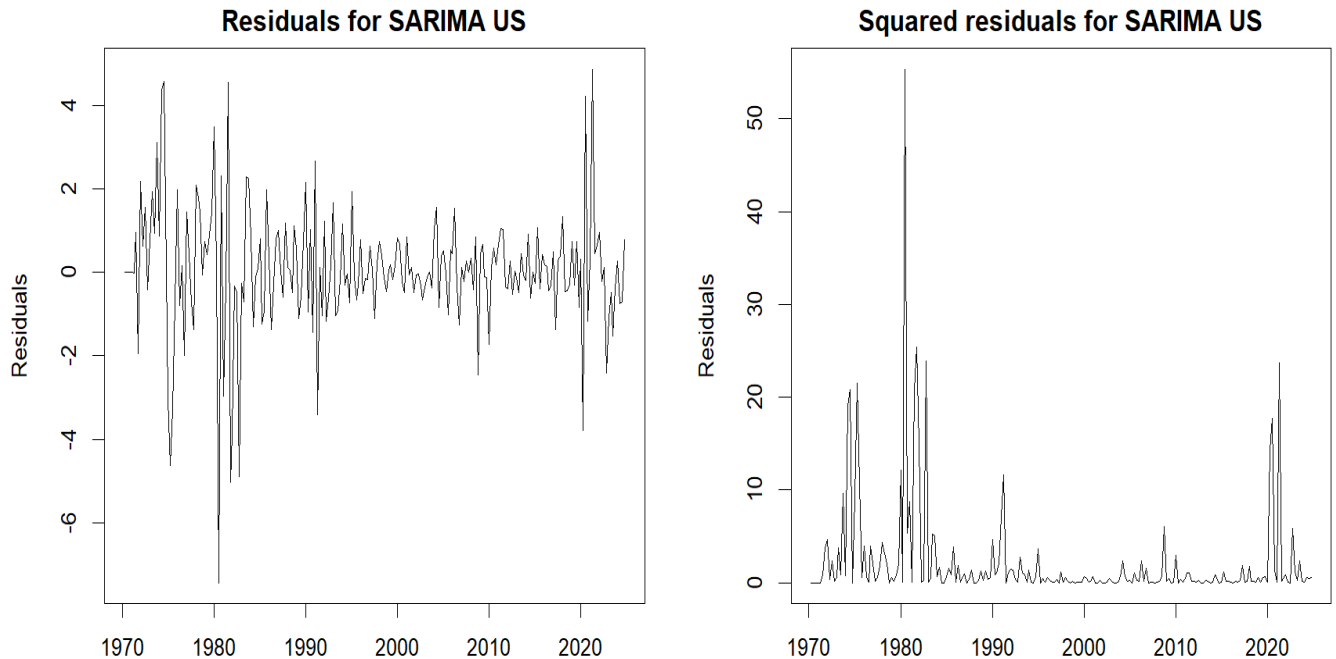
France inflation forecast with GARCH-adjusted intervals





The SARIMA bands are close to the GARCH bands, suggesting that post-Covid volatility is returning towards normal levels, although it remains somewhat higher.

5.2 Foreign inflation : US



Same for US, There is clustering in residuals.

```
> Box.test(ResUS,lag=8) # autocorrelation test on US residuals
Box-Pierce test
data:  ResUS
X-squared = 2.9287, df = 8, p-value = 0.9387
```

There is no remaining autocorrelation in the residuals.

```
> ArchTest(ResUS,8) # ARCH test for US residuals
ARCH LM-test; Null hypothesis: no ARCH effects
data:  ResUS
Chi-squared = 41.494, df = 8, p-value = 0.000001685
> Box.test(ResUS2,lag=8) # autocorrelation test on squared US residuals
Box-Pierce test
data:  ResUS2
X-squared = 68.935, df = 8, p-value = 8.005e-12
```

Both tests confirm the presence of ARCH effects in the squared residuals. Following AIC, we estimate a GARCH(3,1) model.

```
> ##### ESTIMATION
```

```
fitUS=garchFit(ResUS~garch(3,1),data=ResUS,include.mean=FALSE,trace=FALSE) # estimate GARCH for
```

```
Coefficient(s):
```

```
      omega      alpha1      alpha2      alpha3
0.14822990  0.29276961  0.14789232  0.00000001
beta1
0.54994118
```

```
Std. Errors:
based on Hessian
```

```
Error Analysis:
```

```
Estimate Std. Error t value Pr(>|t|)
omega 0.14822990 0.11925713 1.243 0.2139
alpha1 0.29276961 0.12522069 2.338 0.0194 *
alpha2 0.14789232 0.12723689 1.162 0.2451
alpha3 0.00000001 0.20173314 0.000 1.0000
beta1 0.54994118 0.25869128 2.126 0.0335 *
---
```

```
Signif. codes:
```

```
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Standardised Residuals Tests:
```

			Statistic	p-Value
Jarque-Bera Test	R	Chi ²	55.3147855	9.739987e-13
Shapiro-Wilk Test	R	W	0.9653367	3.432797e-05
Ljung-Box Test	R	Q(10)	20.0384917	2.889072e-02
Ljung-Box Test	R	Q(15)	22.8259535	8.790574e-02
Ljung-Box Test	R	Q(20)	39.2119476	6.274150e-03
Ljung-Box Test	R ²	Q(10)	6.0489040	8.111379e-01
Ljung-Box Test	R ²	Q(15)	8.5110576	9.016656e-01
Ljung-Box Test	R ²	Q(20)	10.5846917	9.562902e-01
LM Arch Test	R	TR ²	6.3007186	9.001705e-01

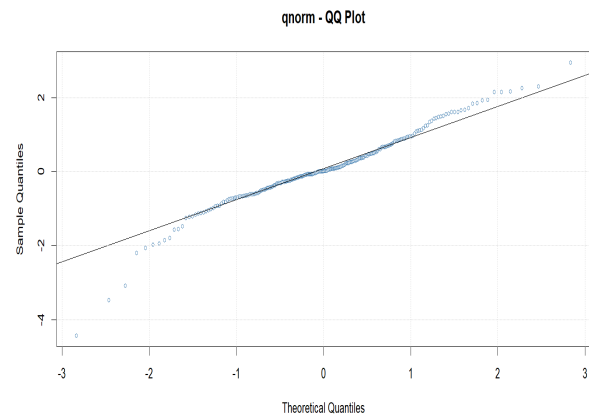
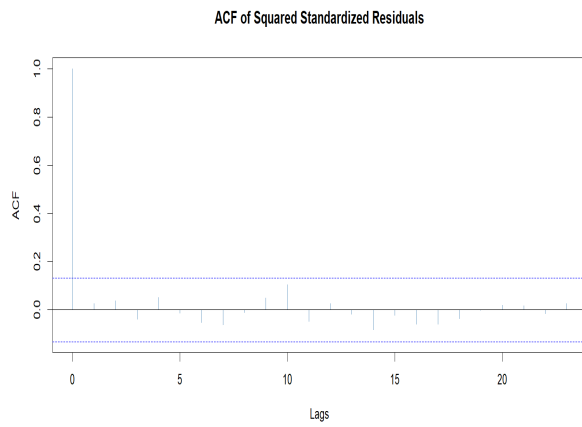
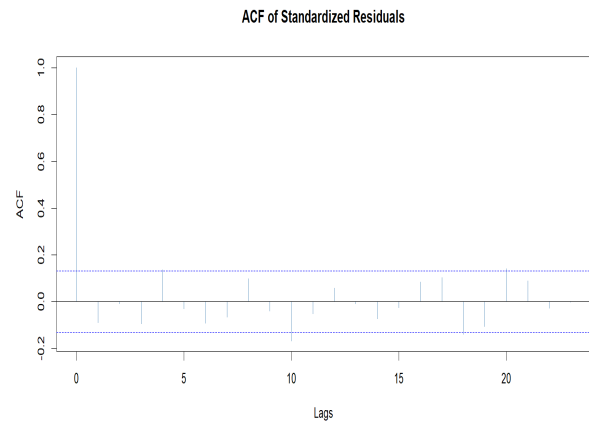
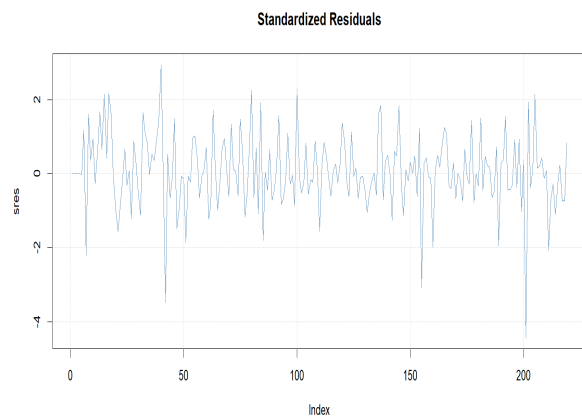
```
Information Criterion Statistics:
```

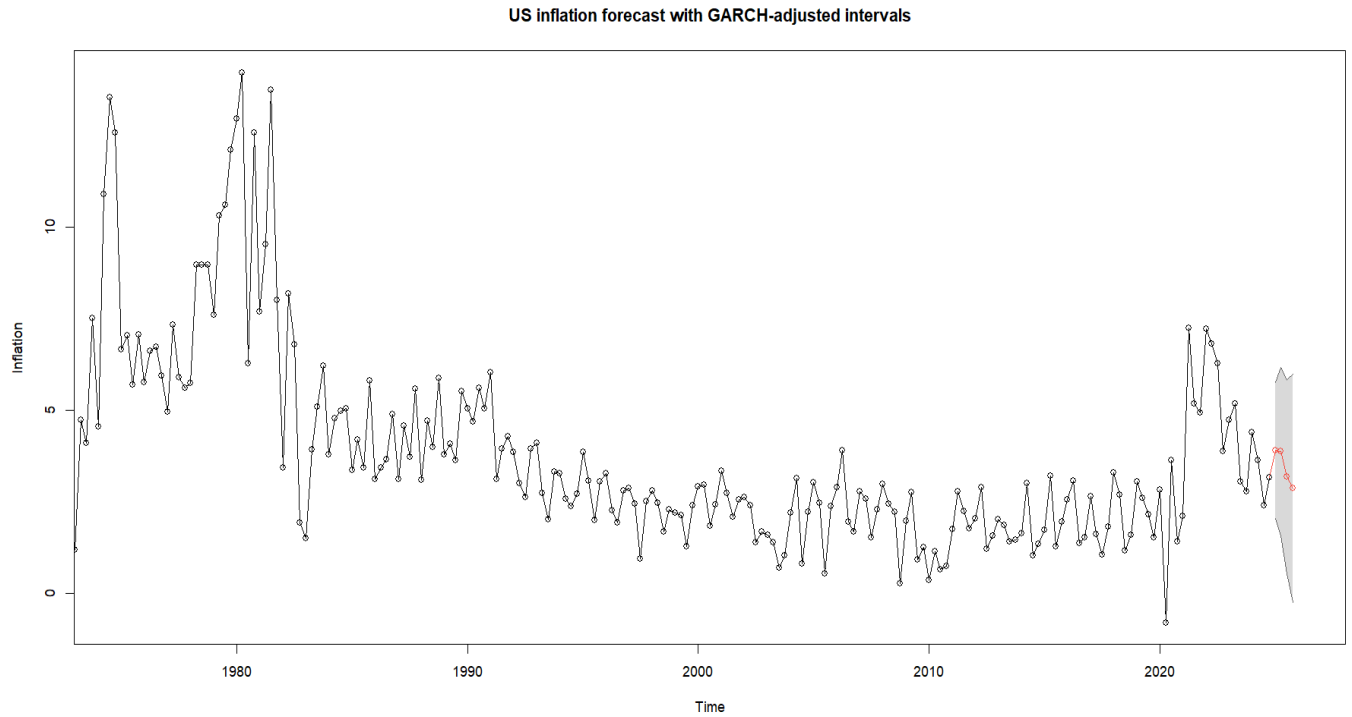
```
      AIC      BIC      SIC      HQIC
3.233156 3.310532 3.232145 3.264406
```

The Ljung–Box tests and the LM ARCH test do not indicate the presence of ARCH effects.

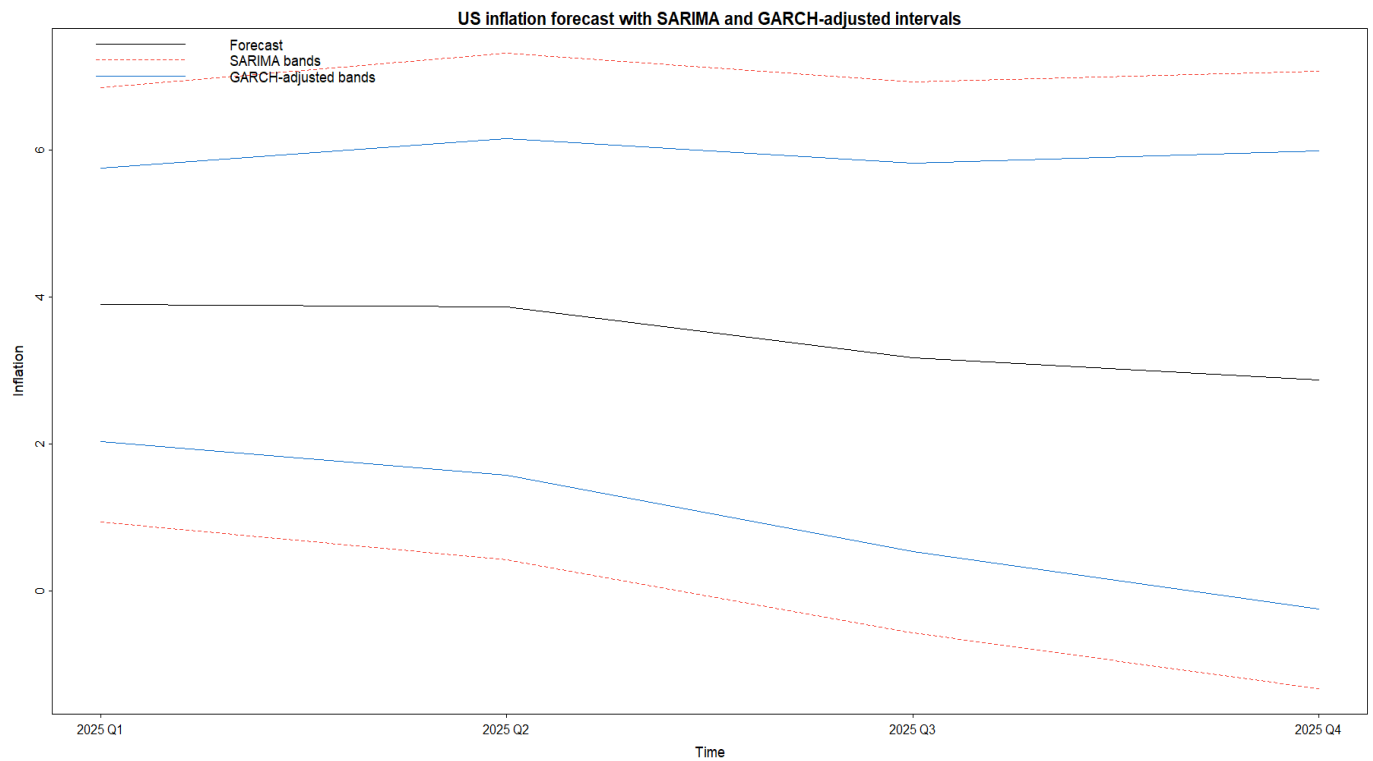
Only Ljung-Box residuals at Q(20) rejects the null.

Residuals diagnostic plots





The downward phase after Covid has lower volatility. Accounting for this, the GARCH model produces narrower prediction bands than the SARIMA.



```
##### QUESTION 5
library(FinTS) # ARCH tests
library(fGarch) # GARCH estimation

##### CONDITIONAL HETEROSCEDASTICITY : HOME COUNTRY FRANCE
m=sarima(INFFR,7,1,1,3,1,1,4) # estimate SARIMA model for France
ResFR=m$fit$residuals # extract residuals
par(mfrow=c(1,2)) # two plots side by side
plot.ts(ResFR,ylab="Residuals",main="Residuals for SARIMA France") #
  residuals
plot.ts(ResFR^2,ylab="Residuals",main="Squared residuals for SARIMA France
  ") # squared residuals
Box.test(ResFR,lag=8) # residual autocorrelation test
ResFR2=ResFR^2 # squared residuals
ArchTest(ResFR,8) # Engle ARCH test
Box.test(ResFR2,lag=8) # autocorrelation test on squared residuals

##### SELECT THE BEST ARCH/GARCH MODEL
aic_stats=numeric(3) # store AIC for ARCH orders
bic_stats=numeric(3) # store BIC for ARCH orders
for(p in 1:3) { # loop over ARCH orders
  fit=garchFit(as.formula(paste("~ garch(",p,",0)",sep="")),data=
    ResFR,include.mean=FALSE,trace=FALSE) # estimate
  aic_stats[p]=fit@fit$ics["AIC"] # save AIC
  bic_stats[p]=fit@fit$ics["BIC"] # save BIC
}
aic_stats # display AIC values
bic_stats # display BIC values
which.min(aic_stats) # best ARCH order by AIC 3
which.min(bic_stats) # best ARCH order by BIC 2

aic_stats=numeric(3) # reset AIC values
bic_stats=numeric(3) # reset BIC values
for(q in 0:2) { # loop over q with p fixed at 2
  fit=garchFit(as.formula(paste("~ garch(2,\",q,\",)\",sep="")),data=
    ResFR,include.mean=FALSE,trace=FALSE) # estimate
  aic_stats[q+1]=fit@fit$ics["AIC"] # save AIC
  bic_stats[q+1]=fit@fit$ics["BIC"] # save BIC
}
aic_stats # display AIC values
bic_stats # display BIC values
which.min(aic_stats)-1 # best q by AIC 1
which.min(bic_stats)-1 # best q by BIC 1

##### ESTIMATION
fitFR=garchFit(ResFR~garch(2,1),data=ResFR,include.mean=FALSE,trace=FALSE)
  # estimate ARCH(2) for France residuals
options(scipen = 3)
summary(fitFR) # ARCH model summary
plot(fitFR) # ARCH model diagnostics
0

##### SARIMA + GARCH BANDS
```

```

fcsar=sarima.for(INFFR,4,7,1,1,3,1,1,4) # 4-step ahead SARIMA forecast
Garchvol=predict(fitFR,n.ahead=4) # forecast GARCH volatility
sdGarch=as.numeric(Garchvol$standardDeviation) # extract forecasted
  standard deviations
sig0=sd(na.omit(ResFR)) # residual standard deviation from SARIMA
adj=sdGarch/sig0 # volatility adjustment factor
lower_sarima=fcsar$pred-1.96*fcsar$se # lower SARIMA 95% band
upper_sarima=fcsar$pred+1.96*fcsar$se # upper SARIMA 95% band
lower_garch=fcsar$pred-1.96*fcsar$se*adj # lower GARCH-adjusted band
upper_garch=fcsar$pred+1.96*fcsar$se*adj # upper GARCH-adjusted band
forecast_table=cbind(
forecast=as.numeric(fcsar$pred), # point forecast
se_sarima=as.numeric(fcsar$se), # SARIMA standard error
sigma_garch=sdGarch, # GARCH volatility forecast
lower95_sarima=as.numeric(lower_sarima), # lower SARIMA interval
upper95_sarima=as.numeric(upper_sarima), # upper SARIMA interval
lower95_garch=as.numeric(lower_garch), # lower adjusted interval
upper95_garch=as.numeric(upper_garch) # upper adjusted interval
)
forecast_table # display forecast table

##### PLOT MODIFIED BANDS
forecast_ts=ts(forecast_table[, "forecast"], start=c(2025,1), frequency=4) #
  forecast series
lower_sar_ts=ts(forecast_table[, "lower95_sarima"], start=c(2025,1),
  frequency=4) # SARIMA lower band
upper_sar_ts=ts(forecast_table[, "upper95_sarima"], start=c(2025,1),
  frequency=4) # SARIMA upper band
lower_garch_ts=ts(forecast_table[, "lower95_garch"], start=c(2025,1),
  frequency=4) # GARCH lower band
upper_garch_ts=ts(forecast_table[, "upper95_garch"], start=c(2025,1),
  frequency=4) # GARCH upper band
a=1:4 # forecast horizons
ylim_all=range(c(forecast_ts, lower_sar_ts, upper_sar_ts, lower_garch_ts,
  upper_garch_ts)) # common y-axis range
plot(a, as.numeric(forecast_ts), type="l", col=1, lty=1, ylim=ylim_all, xaxt="n"
, xlab="Time", ylab="Inflation", main="SARIMA forecast with SARIMA and
  GARCH-adjusted intervals") # forecast path only
0
lines(a, as.numeric(lower_sar_ts), col=2, lty=2) # lower SARIMA band
lines(a, as.numeric(upper_sar_ts), col=2, lty=2) # upper SARIMA band
lines(a, as.numeric(lower_garch_ts), col=4, lty=1) # lower GARCH-adjusted
  band
lines(a, as.numeric(upper_garch_ts), col=4, lty=1) # upper GARCH-adjusted
  band
axis(1, at=a, labels=c("2025 Q1", "2025 Q2", "2025 Q3", "2025 Q4")) # label
  forecast quarters
legend("topleft", legend=c("Forecast", "SARIMA bands", "GARCH-adjusted bands"
), col=c(1,2,4), lty=c(1,2,1), bty="n") # add legend

##### PLOT THE FULL GRAPH
graphics.off() # clear previous plots
library(rgl) # 3D plotting library
plot(INFFR, pch=1, type="o", xlim=c(1975,2026), ylim=range(INFFR, lower_garch_

```

```

    ts, upper_garch_ts), xlab="Time", ylab="Inflation", main="France inflation
    forecast with GARCH-adjusted intervals") # full historical plot
polygon(c(time(forecast_ts), rev(time(forecast_ts))), c(as.numeric(upper_
    garch_ts), rev(as.numeric(lower_garch_ts))), col=gray(0.85), border=NA) #
    shaded adjusted interval
lines(c(time(INFFR)[length(INFFR)], time(forecast_ts)), c(tail(INFFR,1), as.
    numeric(forecast_ts)), col=2, type="l") # connect last observation to
    forecast
points(forecast_ts, col=2, pch=1) # forecast points
points(time(INFFR)[length(INFFR)], tail(INFFR,1), col=1, pch=1) # last
    observed point
lines(lower_garch_ts, col="gray40") # lower adjusted band
lines(upper_garch_ts, col="gray40") # upper adjusted band

##### CONDITIONAL HETEROSCEDASTICITY : FOREIGN COUNTRY US
par(mfrow=c(1,1), mar=c(3,4,2,2), oma=c(0,0,2,0)) # single plot window
acf2(dINFUSS) # ACF and PACF for seasonal differenced US inflation
mUS=sarima(INFUS,2,1,0,0,1,1,4) # SARIMA model for US inflation
ResUS=mUS$fit$residuals # residuals from US SARIMA model
par(mfrow=c(1,2)) # two plots side by side
plot.ts(ResUS, ylab="Residuals", main="Residuals for SARIMA US") # US
    residuals
plot.ts(ResUS^2, ylab="Residuals", main="Squared residuals for SARIMA US") #
    US squared residuals
Box.test(ResUS, lag=8) # autocorrelation test on US residuals
ResUS2=ResUS^2 # squared US residuals
ArchTest(ResUS, 8) # ARCH test for US residuals
Box.test(ResUS2, lag=8) # autocorrelation test on squared US residuals
##### SELECT THE BEST ARCH/GARCH MODEL FOR US
aic_stats=numeric(3) # store AIC for ARCH orders
bic_stats=numeric(3) # store BIC for ARCH orders
for(p in 1:3) { # loop over ARCH orders
    fit=garchFit(as.formula(paste("~ garch(", p, ",0)", sep="")),
        data=ResUS, include.mean=FALSE, trace=FALSE) # estimate
    aic_stats[p]=fit@fit$ics["AIC"] # save AIC
    bic_stats[p]=fit@fit$ics["BIC"] # save BIC
}
aic_stats # display AIC values
bic_stats # display BIC values
which.min(aic_stats) # best ARCH order by AIC
which.min(bic_stats) # best ARCH order by BIC

aic_stats=numeric(3) # reset AIC values
bic_stats=numeric(3) # reset BIC values
for(q in 0:2) { # loop over q with p fixed at 2
    fit=garchFit(as.formula(paste("~ garch(3, ", q, ")", sep="")),
        data=ResUS, include.mean=FALSE, trace=FALSE) # estimate
    aic_stats[q+1]=fit@fit$ics["AIC"] # save AIC
    bic_stats[q+1]=fit@fit$ics["BIC"] # save BIC
}
aic_stats # display AIC values
bic_stats # display BIC values
which.min(aic_stats)-1 # best q by AIC
which.min(bic_stats)-1 # best q by BIC

```

```
##### ESTIMATION
fitUS=garchFit(ResUS~garch(3,1),data=ResUS,include.mean=FALSE,trace=FALSE)
# estimate GARCH for US residuals
options(scipen=3)
summary(fitUS) # GARCH model summary
plot(fitUS) # GARCH model diagnostics
0
##### SARIMA + GARCH BANDS
fcsarus=sarima.for(INFUS,4,2,1,0,0,1,1,4) # 4-step ahead SARIMA forecast
for US
Garchvol=predict(fitUS,n.ahead=4) # forecast GARCH volatility
sdGarch=as.numeric(Garchvol$standardDeviation) # extract forecasted
standard deviations
sig0=sd(na.omit(ResUS)) # residual standard deviation from SARIMA
adj=sdGarch/sig0 # volatility adjustment factor
lower_sarima=fcsarus$pred-1.96*fcsarus$se # lower SARIMA 95% band
upper_sarima=fcsarus$pred+1.96*fcsarus$se # upper SARIMA 95% band
lower_garch=fcsarus$pred-1.96*fcsarus$se*adj # lower GARCH-adjusted band
upper_garch=fcsarus$pred+1.96*fcsarus$se*adj # upper GARCH-adjusted band
forecast_table=cbind(
forecast=as.numeric(fcsarus$pred), # point forecast
se_sarima=as.numeric(fcsarus$se), # SARIMA standard error
sigma_garch=sdGarch, # GARCH volatility forecast
lower95_sarima=as.numeric(lower_sarima), # lower SARIMA interval
upper95_sarima=as.numeric(upper_sarima), # upper SARIMA interval
lower95_garch=as.numeric(lower_garch), # lower adjusted interval
upper95_garch=as.numeric(upper_garch) # upper adjusted interval
)
forecast_table # display forecast table

##### PLOT MODIFIED BANDS
forecast_ts=ts(forecast_table[, "forecast"], start=c(2025,1), frequency=4) #
forecast series
lower_sar_ts=ts(forecast_table[, "lower95_sarima"], start=c(2025,1),
frequency=4) # SARIMA lower band
upper_sar_ts=ts(forecast_table[, "upper95_sarima"], start=c(2025,1),
frequency=4) # SARIMA upper band
lower_garch_ts=ts(forecast_table[, "lower95_garch"], start=c(2025,1),
frequency=4) # GARCH lower band
upper_garch_ts=ts(forecast_table[, "upper95_garch"], start=c(2025,1),
frequency=4) # GARCH upper band
a=1:4 # forecast horizons
ylim_all=range(c(forecast_ts, lower_sar_ts, upper_sar_ts, lower_garch_ts,
upper_garch_ts)) # common y-axis range
plot(a, as.numeric(forecast_ts), type="l", col=1, lty=1, ylim=ylim_all, xaxt="n"
,
xlab="Time", ylab="Inflation",
main="US inflation forecast with SARIMA and GARCH-adjusted intervals") #
forecast path only
lines(a, as.numeric(lower_sar_ts), col=2, lty=2) # lower SARIMA band
lines(a, as.numeric(upper_sar_ts), col=2, lty=2) # upper SARIMA band
lines(a, as.numeric(lower_garch_ts), col=4, lty=1) # lower GARCH-adjusted
band
```

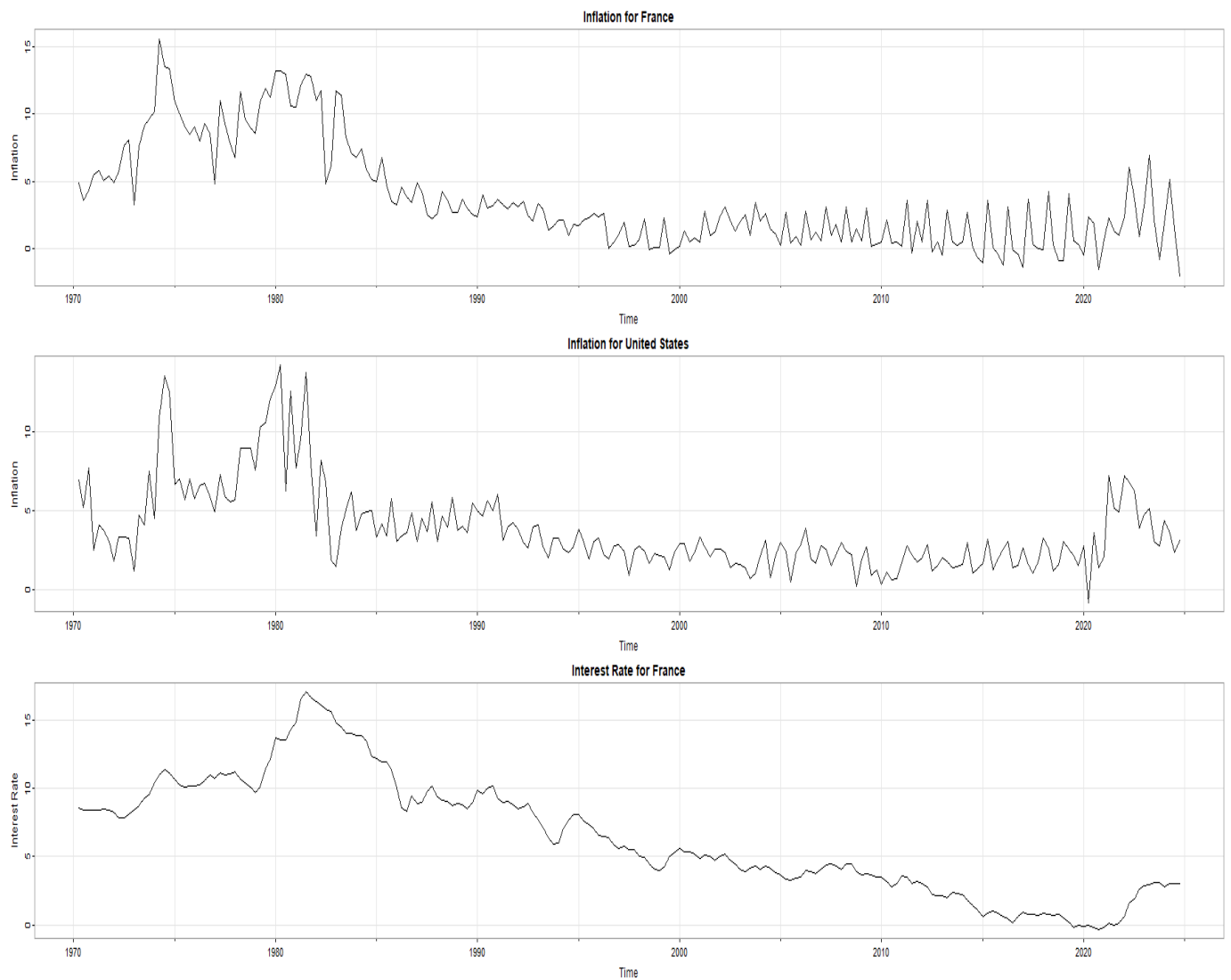
```

lines(a,as.numeric(upper_garch_ts),col=4,lty=1) # upper GARCH-adjusted
band
axis(1,at=a,labels=c("2025 Q1","2025 Q2","2025 Q3","2025 Q4")) # label
forecast quarters
legend("topleft",legend=c("Forecast","SARIMA bands","GARCH-adjusted bands"
),
col=c(1,2,4),lty=c(1,2,1),bty="n") # add legend

##### PLOT THE FULL GRAPH
graphics.off() # clear previous plots
plot(INFUS,pch=1,type="o",xlim=c(1975,2026),
ylim=range(INFUS,lower_garch_ts,upper_garch_ts),
xlab="Time",ylab="Inflation",
main="US inflation forecast with GARCH-adjusted intervals") # full
historical plot
polygon(c(time(forecast_ts),rev(time(forecast_ts))),
c(as.numeric(upper_garch_ts),rev(as.numeric(lower_garch_ts))),
col=gray(0.85),border=NA) # shaded adjusted interval
lines(c(time(INFUS)[length(INFUS)],time(forecast_ts)),
c(tail(INFUS,1),as.numeric(forecast_ts)),
col=2,type="l") # connect last observation to forecast
points(forecast_ts,col=2,pch=1) # forecast points
points(time(INFUS)[length(INFUS)],tail(INFUS,1),col=1,pch=1) # last
observed point
lines(lower_garch_ts,col="gray40") # lower adjusted band
lines(upper_garch_ts,col="gray40") # upper adjusted band

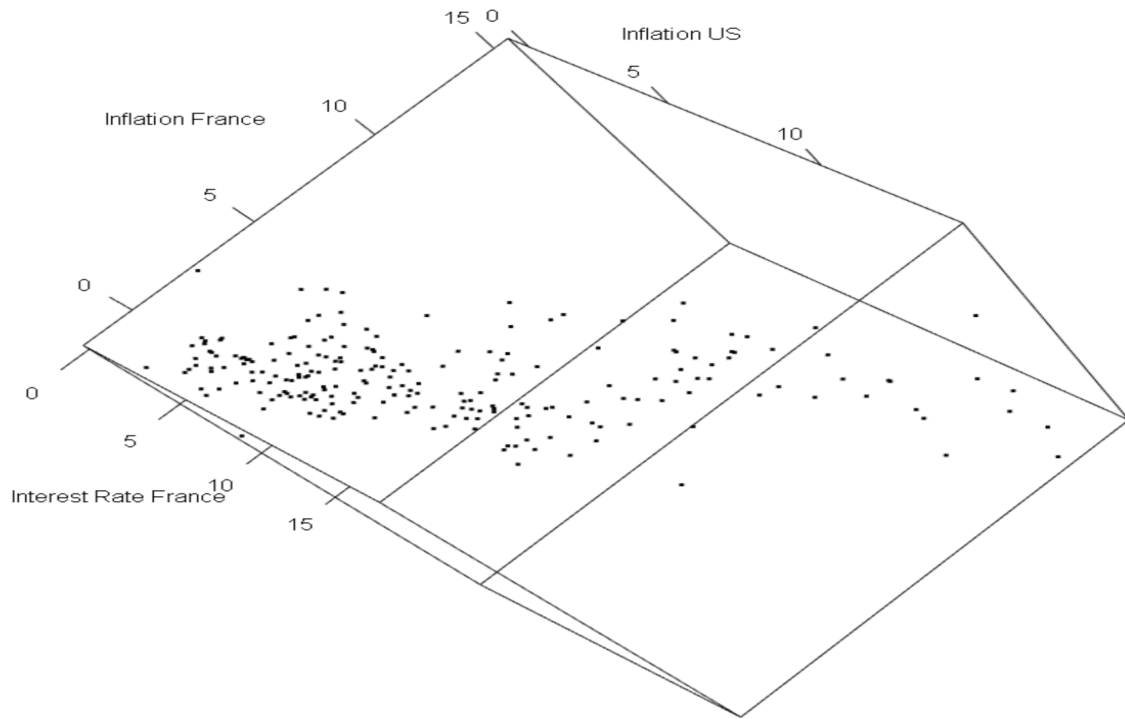
```

6 Cointegrated VAR



The high values during periods of stress reflect changing economic conditions rather than a clear deterministic trend. Inflation and interest rates values generally remain within a stable range over time. Therefore, we estimate a model with a constant.

Figure 3: 3D Graphical analysis.



The 3D plot suggests long-run comovement among the variables. Knowing, all three variables are $I(1)$, cointegration is possible.

Table 4: Johansen trace test results

Johansen Trace Test				
Hypothesis	Test Stat	10%	5%	1%
$r \leq 2$	2.36	7.52	9.24	12.97
$r \leq 1$	11.30	17.85	19.96	24.60
$r = 0$	47.05	32.00	34.91	41.07
Cointegration Vectors (normalised)				
Variable	INFFR.I5	INFUS.I5	IR10FR.I5	Constant
INFFR.I5	1.000	1.000	1.000	1.000
INFUS.I5	-5.741	-0.775	-0.477	0.314
IR10FR.I5	1.496	-0.401	4.610	-2.243
Constant	9.403	1.733	-25.889	30.632
Loading Matrix (Adjustment Coefficients)				
	INFFR.I5	INFUS.I5	IR10FR.I5	
Δ INFFR	0.022	-0.062	-0.006	
Δ INFUS	0.049	0.150	-0.003	
Δ IR10FR	-0.015	0.029	-0.001	

The trace test rejects $r = 0$ at the 5% level but does not reject $r \leq 1$, indicating a cointegration rank of 1.

The estimated cointegrating vector is:

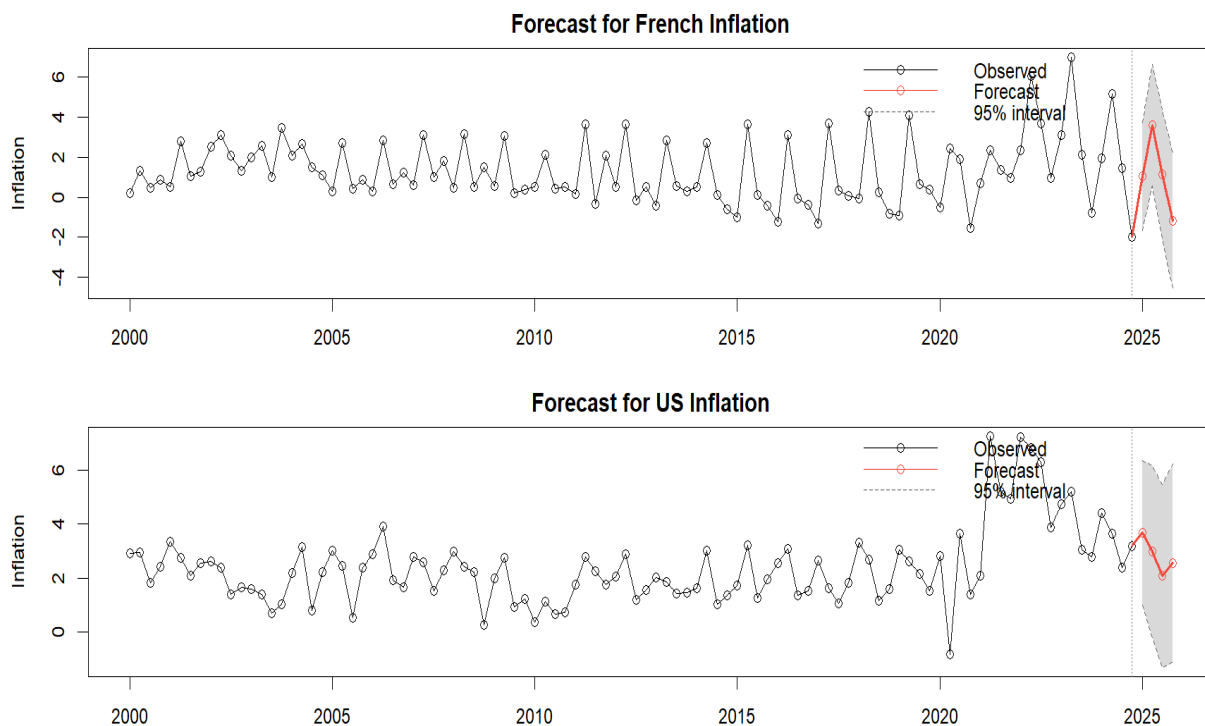
$$INFFR - 5.741 INFUS + 1.496 IR10FR + 9.403 = 0$$

or

$$INFFR = 5.741 INFUS - 1.496 IR10FR - 9.403.$$

The normalized cointegrating relation suggests that, in the long run, French inflation is positively associated with US inflation and negatively associated with the French interest rate.

The adjustment coefficients suggest weak adjustment in French inflation (0.022), somewhat stronger adjustment in US inflation (0.049).



The forecast appears reasonable, as it preserves the same downward trend pattern.

The appendix presents the AIC results, VAR tests, residual plots, and the Johansen maximum eigenvalue test.

```
##### QUESTION 6
library(vars)
library("astsa")
library("FinTS")
library("fGarch")
library("rgl")
library("urca")
```

```

library("tsDyn")
par(mfrow=c(3,1),mar=c(3,4,2,2),oma=c(0,0,2,0)) # layout for series plots
tsplot(INFFR,col=couleur,ylab="Inflation",xlab="Time",main="Inflation for
  France") # France inflation
tsplot(INFUS,col=couleur,ylab="Inflation",xlab="Time",main="Inflation for
  United States") # US inflation
tsplot(IR10FR,col=couleur,ylab="Interest Rate",xlab="Time",main="Interest
  Rate for France") # France interest rate
plot3d(INFFR,INFUS,IR10FR,ylab="Inflation US",xlab="Inflation France",zlab
  ="Interest Rate France") # 3D plot of all variables

VARselect(x,lag.max=7,type="const") # select lag length for VAR
var5=VAR(x,5,type="const") # estimate VAR with lags
u_hat=ts(residuals(var5)) # extract VAR residuals
plot.ts(u_hat,main="Residuals") # inspect residuals
serial.test(var5,lags.pt=7) # Portmanteau test for residual serial
  correlation
serial.test(var5,lags.bg=7) # Breusch-Godfrey serial correlation test
arch.test(var5,lags.multi=7) # multivariate ARCH test
normality.test(var5) # residual normality test
cvar.t=ca.jo(x,type="trace",ecdet="const",K=5) # Johansen trace test
summary(cvar.t) # trace test summary
cvar.e=ca.jo(x,type="eigen",ecdet="const",K=5) # Johansen maximum
  eigenvalue test
summary(cvar.e) # max-eigen test summary
vecm_var=vec2var(cvar.e,r=1) # convert cointegrated system to VAR form
forecast=predict(vecm_var,n.ahead=4,ci=0.95) # 5-step ahead forecast

plot_fc=function(series,name,title) { # function to plot one forecast
  series
    f=frequency(series) # frequency of the observed series
    s=if(end(series)[2]<f) c(end(series)[1],end(series)[2]+1) else c(
  end(series)[1]+1,1) # next period
    fc=ts(forecast$fcst[[name]][,1],start=s,frequency=f) # forecast
  mean
    lo=ts(forecast$fcst[[name]][,2],start=s,frequency=f) # lower bound
    hi=ts(forecast$fcst[[name]][,3],start=s,frequency=f) # upper bound
    obs=window(series,start=c(2000,1)) # observed sample shown on
  graph
    plot(obs,type="o",pch=1,xlim=c(2000,tail(time(hi),1)),ylim=range(
  obs,lo,hi,na.rm=TRUE),xlab="Time",ylab="Inflation",main=title) #
  observed series
    polygon(c(time(fc),rev(time(fc))),c(as.numeric(hi),rev(as.numeric(
  lo))),col=gray(0.85),border=NA) # shaded forecast interval
    lines(obs,col=1) # observed path on top
    lines(c(tail(time(series),1),time(fc)),c(tail(series,1),as.numeric
  (fc)),col=2,lwd=2) # connect last observed point to forecast
    lines(lo,col="gray40",lty=2) # lower interval bound
    lines(hi,col="gray40",lty=2) # upper interval bound
    points(fc,col=2,pch=1) # forecast points
    abline(v=tail(time(series),1),lty=3,col="gray50") # mark forecast
  origin
    legend("topright",legend=c("Observed","Forecast","95% interval"),
  col=c(1,2,"gray40"),lty=c(1,1,2),pch=c(1,1,NA),bty="n") # legend

```

```

}

graphics.off()
par(mfrow=c(2,1),mar=c(3,4,2,2)) # layout for final inflation forecasts
plot_fc(INFFR,"INFFR","Forecast for French Inflation") # forecast plot for
  France inflation
plot_fc(INFUS,"INFUS","Forecast for US Inflation") # forecast plot for US
  inflation
plot(forecast) # default forecast plot

```

Appendix

Question 3

Bernhard Pfaff procedure for US inflation

```

> ##### UNIT ROOT TESTS : US INFLATION
> summary(ur.df(INFUS,lags=8,type="trend",selectlags="AIC")) # level with trend
Test regression trend
Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-5.7164 -0.6172 -0.1532  0.5591  6.1065
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.409974   0.468592   3.009  0.00296 **
z.lag.1      -0.204435   0.061736  -3.311  0.00110 **
tt           -0.005522   0.002319  -2.381  0.01819 *
z.diff.lag1  -0.297472   0.083368  -3.568  0.00045 ***
z.diff.lag2  -0.208472   0.082436  -2.529  0.01222 *
z.diff.lag3   0.001209   0.083302   0.015  0.98843
z.diff.lag4   0.189341   0.082337   2.300  0.02251 *
z.diff.lag5   0.028587   0.083633   0.342  0.73285
z.diff.lag6   0.066471   0.082164   0.809  0.41948
z.diff.lag7  -0.139543   0.077579  -1.799  0.07358 .
z.diff.lag8   0.110917   0.069268   1.601  0.11090
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.5 on 199 degrees of freedom
Multiple R-squared:  0.3418, Adjusted R-squared:  0.3087
F-statistic: 10.33 on 10 and 199 DF,  p-value: 5.606e-14

Value of test-statistic is: -3.3114 3.6679 5.5016

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2  6.22  4.75  4.07
phi3  8.43  6.49  5.47

```

```
> summary(ur.df(INFUS,lags=8,type="drift",selectlags="AIC")) # level with drift
Test regression drift
Call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-6.0796 -0.5845 -0.1182  0.5633  6.1224
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.37086    0.20304   1.827 0.069247 .
z.lag.1      -0.09691    0.04557  -2.126 0.034685 *
z.diff.lag1 -0.39950    0.07482  -5.339 2.51e-07 ***
z.diff.lag2 -0.27436    0.07957  -3.448 0.000687 ***
z.diff.lag3 -0.05961    0.08115  -0.735 0.463470
z.diff.lag4  0.14907    0.08099   1.841 0.067157 .
z.diff.lag5 -0.03741    0.08092  -0.462 0.644406
z.diff.lag6 -0.01777    0.07610  -0.233 0.815641
z.diff.lag7 -0.22788    0.06748  -3.377 0.000879 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.519 on 201 degrees of freedom
Multiple R-squared:  0.3181, Adjusted R-squared:  0.291
F-statistic: 11.72 on 8 and 201 DF,  p-value: 1.19e-13
Value of test-statistic is: -2.1265 2.261
```

Critical values for test statistics:

```
      1pct  5pct 10pct
tau2 -3.46 -2.88 -2.57
phi1  6.52  4.63  3.81
```

```
> summary(ur.df(INFUS,lags=8,type="none",selectlags="AIC")) # level without deterministic term
Test regression none
Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-6.5020 -0.4345 -0.0105  0.7315  5.7412
Coefficients:
Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.02562    0.02366  -1.083 0.280282
z.diff.lag1 -0.45027    0.06986  -6.445 8.32e-10 ***
z.diff.lag2 -0.31656    0.07658  -4.134 5.22e-05 ***
z.diff.lag3 -0.09500    0.07926  -1.199 0.232061
z.diff.lag4  0.11338    0.07905   1.434 0.153038
z.diff.lag5 -0.07302    0.07899  -0.924 0.356382
z.diff.lag6 -0.04698    0.07483  -0.628 0.530786
z.diff.lag7 -0.24826    0.06693  -3.709 0.000269 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.527 on 202 degrees of freedom

Multiple R-squared: 0.3068, Adjusted R-squared: 0.2794

F-statistic: 11.18 on 8 and 202 DF, p-value: 4.731e-13

Value of test-statistic is: -1.0826

Critical values for test statistics:

1pct 5pct 10pct
tau1 -2.58 -1.95 -1.62

> summary(ur.df(dINFUS,lags=8,type="trend",selectlags="AIC")) # difference with trend

Test regression trend

Call:

lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:

Min	1Q	Median	3Q	Max
-6.7992	-0.5225	-0.0861	0.6436	5.5523

Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept)	0.0198062	0.2261013	0.088	0.93028
z.lag.1	-2.1922640	0.2997977	-7.312	6.18e-12 ***
tt	-0.0001963	0.0017651	-0.111	0.91156
z.diff.lag1	0.7230708	0.2795909	2.586	0.01041 *
z.diff.lag2	0.3917664	0.2507034	1.563	0.11971
z.diff.lag3	0.2861434	0.2176511	1.315	0.19012
z.diff.lag4	0.3895142	0.1728218	2.254	0.02529 *
z.diff.lag5	0.3059504	0.1214127	2.520	0.01252 *
z.diff.lag6	0.2553898	0.0671499	3.803	0.00019 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.539 on 200 degrees of freedom

Multiple R-squared: 0.7458, Adjusted R-squared: 0.7356

F-statistic: 73.35 on 8 and 200 DF, p-value: < 2.2e-16

Value of test-statistic is: -7.3125 17.8254 26.7374

Critical values for test statistics:

1pct 5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2 6.22 4.75 4.07
phi3 8.43 6.49 5.47

```
> summary(ur.df(dINFUS,lags=8,type="none",selectlags="AIC")) # difference without deterministic trend
Test regression none
Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-6.7879 -0.5183 -0.0852  0.6462  5.5321
Coefficients:
Estimate Std. Error t value Pr(>|t|)
z.lag.1      -2.1920     0.2983  -7.348 4.88e-12 ***
z.diff.lag1    0.7229     0.2782   2.599 0.010052 *
z.diff.lag2    0.3918     0.2495   1.571 0.117861
z.diff.lag3    0.2863     0.2166   1.322 0.187659
z.diff.lag4    0.3898     0.1719   2.267 0.024464 *
z.diff.lag5    0.3062     0.1208   2.535 0.012002 *
z.diff.lag6    0.2556     0.0668   3.826 0.000174 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.531 on 202 degrees of freedom
Multiple R-squared:  0.7458, Adjusted R-squared:  0.737
F-statistic: 84.66 on 7 and 202 DF, p-value: < 2.2e-16
```

Value of test-statistic is: -7.3481

Critical values for test statistics:

```
1pct 5pct 10pct
tau1 -2.58 -1.95 -1.62
```

Bernhard Pfaff procedure for Interest rate for France

```
> ##### UNIT ROOT TESTS : FRANCE INTEREST RATE
> summary(ur.df(IR10FR,lags=8,type="trend",selectlags="AIC")) # level with trend
Test regression trend
Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
Residuals:
Min      1Q  Median      3Q      Max
-1.01373 -0.24328 -0.01541  0.20302  1.56798
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.596286   0.201844   2.954  0.00350 **
z.lag.1      -0.042943   0.014088  -3.048  0.00260 **
tt           -0.002936   0.001011  -2.905  0.00407 **
z.diff.lag    0.413663   0.062524   6.616 3.12e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 0.3921 on 206 degrees of freedom
Multiple R-squared:  0.1974, Adjusted R-squared:  0.1857
```

F-statistic: 16.89 on 3 and 206 DF, p-value: 7.605e-10

Value of test-statistic is: -3.0481 3.224 4.7199

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

```
> summary(ur.df(IR10FR,lags=8,type="drift",selectlags="AIC")) # level with drift
```

Test regression drift

Call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.05268	-0.24003	-0.02469	0.20707	1.46793

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.027027	0.049229	0.549 0.584
z.lag.1	-0.006227	0.006335	-0.983 0.327
z.diff.lag	0.403878	0.063545	6.356 1.3e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.399 on 207 degrees of freedom

Multiple R-squared: 0.1645, Adjusted R-squared: 0.1564

F-statistic: 20.38 on 2 and 207 DF, p-value: 8.348e-09

Value of test-statistic is: -0.9831 0.5953

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.46	-2.88	-2.57
phi1	6.52	4.63	3.81

```
> summary(ur.df(IR10FR,lags=8,type="none",selectlags="AIC")) # level without deterministic term
```

Test regression none

Call:

```
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.06003	-0.23056	-0.01192	0.23236	1.45383

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
z.lag.1	-0.003347	0.003543	-0.945 0.346
z.diff.lag	0.401002	0.063223	6.343 1.38e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 0.3984 on 208 degrees of freedom
Multiple R-squared: 0.1657, Adjusted R-squared: 0.1576
F-statistic: 20.65 on 2 and 208 DF, p-value: 6.608e-09

Value of test-statistic is: -0.9445

Critical values for test statistics:

1pct 5pct 10pct
tau1 -2.58 -1.95 -1.62

```
> summary(ur.df(dIR10FR,lags=8,type="trend",selectlags="AIC")) # difference with trend
Test regression trend
```

Call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.08160	-0.23057	-0.01414	0.22386	1.43032

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0041617	0.0589262	0.071 0.944
z.lag.1	-0.6299567	0.0763276	-8.253 1.86e-14 ***
tt	-0.0001642	0.0004604	-0.357 0.722
z.diff.lag	0.0505405	0.0696194	0.726 0.469

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 0.4012 on 205 degrees of freedom
Multiple R-squared: 0.3016, Adjusted R-squared: 0.2914
F-statistic: 29.51 on 3 and 205 DF, p-value: 6.67e-16

Value of test-statistic is: -8.2533 22.7063 34.0589

Critical values for test statistics:

1pct 5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2 6.22 4.75 4.07
phi3 8.43 6.49 5.47

```
> summary(ur.df(dIR10FR,lags=8,type="none",selectlags="AIC")) # difference without determinist.
Test regression none
```

Call:

```
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.08655	-0.25442	-0.03462	0.20266	1.42504

```

Coefficients:
Estimate Std. Error t value Pr(>|t|)
z.lag.1    -0.62624    0.07580  -8.262  1.7e-14 ***
z.diff.lag  0.04873    0.06929   0.703   0.483
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 0.3997 on 207 degrees of freedom
Multiple R-squared:  0.3003, Adjusted R-squared:  0.2935
F-statistic: 44.41 on 2 and 207 DF,  p-value: < 2.2e-16

```

Value of test-statistic is: -8.2616

```

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

Question 6

selection of lags

```

> VARselect(x,lag.max=7,type="const") # select lag length for VAR
$selection
AIC(n)  HQ(n)  SC(n) FPE(n)
5       5      4       5

$criteria
      1      2      3      4      5      6      7
AIC(n) 0.2116772 0.05734693 0.01907711 -0.3995248 -0.5381598 -0.5002676 -0.49574154
HQ(n)  0.2884689 0.19173245 0.21105641 -0.1499517 -0.2309929 -0.1355069 -0.07338706
SC(n)  0.4016727 0.38983897 0.49406573  0.2179604  0.2218220  0.4022108  0.54923344
FPE(n) 1.2357656 1.05909945 1.01947450  0.6709487  0.5843255  0.6072462  0.61049088

```

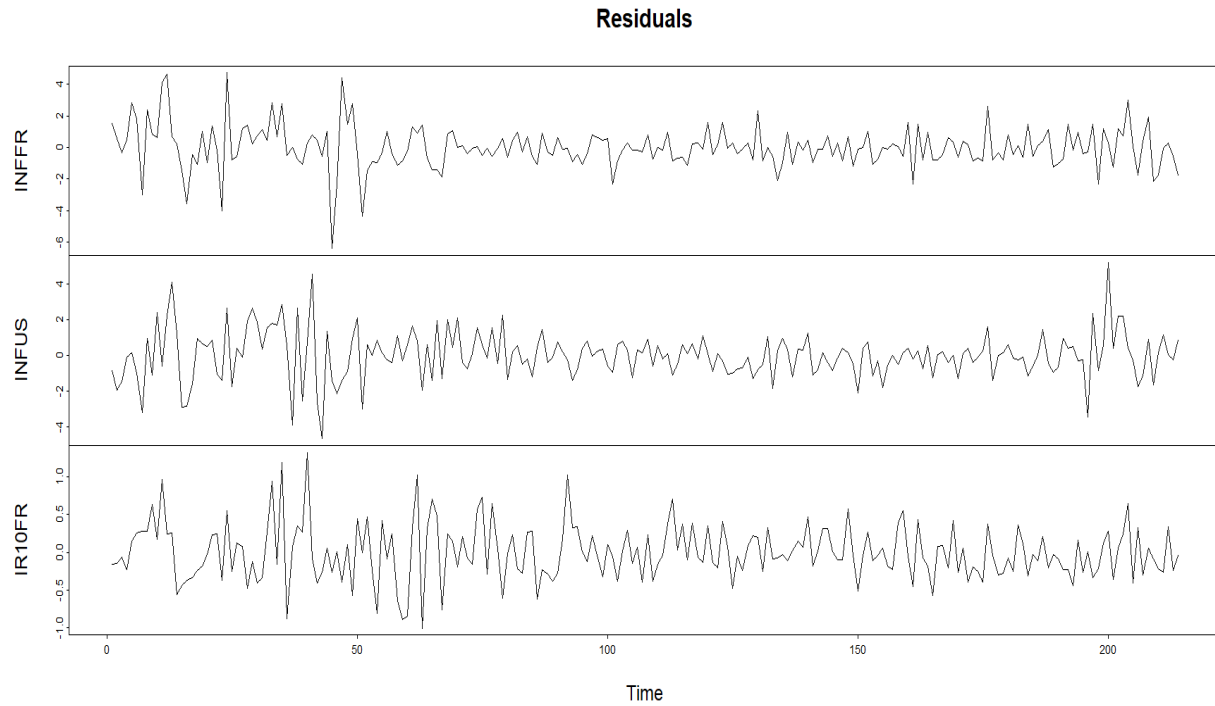


Table 5: Residual diagnostics for the VAR(5) model

Test	Statistic	p-value	Conclusion
Portmanteau test	40.157	0.002	Reject no residual serial correlation
Breusch–Godfrey test	128.520	0.025	Reject no residual serial correlation
Multivariate ARCH test	515.950	< 0.001	Reject no ARCH effects
Jarque–Bera normality test	185.750	< 0.001	Reject multivariate normality
Skewness test	3.516	0.319	Do not reject symmetric residuals
Kurtosis test	182.240	< 0.001	Reject normal kurtosis

Table 6: Johansen maximum eigenvalue test results

Johansen Maximum Eigenvalue Test				
Hypothesis	Test Stat	10%	5%	1%
$r \leq 2$	2.36	7.52	9.24	12.97
$r \leq 1$	8.93	13.75	15.67	20.20
$r = 0$	35.75	19.77	22.00	26.81
Cointegration Vectors (normalised)				
Variable	INFFR.I5	INFUS.I5	IR10FR.I5	Constant
INFFR.I5	1.000	1.000	1.000	1.000
INFUS.I5	-5.741	-0.775	-0.477	0.314
IR10FR.I5	1.496	-0.401	4.610	-2.243
Constant	9.403	1.733	-25.889	30.632
Loading Matrix (Adjustment Coefficients)				
	INFFR.I5	INFUS.I5	IR10FR.I5	
Δ INFFR	0.022	-0.062	-0.006	
Δ INFUS	0.049	0.150	-0.003	
Δ IR10FR	-0.015	0.029	-0.001	